TS5000 Robot Controller

Controller Manual

(Compatible Robots: THE600，THE800，THE1000)

Rev.4

October 1, 2022

Important

Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
Before operating the industrial robot, read through and completely understand this manual.
After reading through this manual, keep it nearby for future reference.

Shibaura Machine

This instruction manual applies to the controller TS5000.

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Introduction

Thank you very much for your purchase of SHIBAURA MACHINE robot system.
This manual describes the standard specifications, installation, wiring, and maintenance of the controller. Read this manual before unpacking the product.
Before using this manual, be sure to read a separate instruction manual STEA1179_THE Series Safety Manual to ensure that you fully understand the safety measures.

This manual has the following chapters.

Chapter 1: Specifications
This chapter describes the standard specifications of the controller.

Chapters 2 to 5: Installation
These chapters describe transportation, unpacking, carrying and installation of the controller.
These chapters also describe the installation environment, required space, and installation procedure for the controller, and wiring between the robot and the controller and between the controller and peripheral devices.

Chapters 6 to 8: Maintenance
These chapters describe maintenance of the controller.

Chapters 9: disposal
These chapters describe disposal of the controller.

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Use Range

The products have been developed with intention of adapting it to automation of wide range of work including conveying work such as movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should use these products with the aim to adapt it to automation of these work and incorporate it to the automation equipment.

This product is not intended to apply it to work that contacts a human (such as work related to medical care and nursing care), incorporate it to the equipment that supports human life (such as life support device and incubator), incorporate it to the equipment on which a human rides (including transport equipment such as car, aerospace equipment such as airplane, and game equipment such as roller coaster), and incorporate it to the equipment that protects human life and human body (such as safety device).
This product is not intended to incorporate it to the equipment aimed to control large quantity of heat (such as nuclear power controller and incinerator controller).
This product is prohibited from being used to apply it to work or incorporate it to the equipment that is likely to threaten human life and harm a human body directly due to a failure or malfunction. If this product is used, it will be guaranteed under customer's responsibility.

Use Method

This describes the operating conditions for the series. Usage under different conditions could expose the user to unexpected risks. Be sure to follow the required conditions.

1. Main specifications of the product Refer to the robot manual for your robot.
2. Restrictions on use: Tables 1 and 2 give "Restricted Specification Sheet" indicating restrictions on use.

Table 1 Restricted Specification Sheet (1)

1. Purpose and application of using product (restrictions on use)	
1) Intended use	Allow the robot to achieve operation along with the purposes (article conveyance, assembly, inspection, machining, etc.). (1) During teaching and adjustment operation - Operation is likely to be performed by turning on the servo power within the movable range of the robot. - A person who observes teaching and operation checking worker as a co-worker is arranged outside the movable range. (1) Teach the position where the robot operates. (2) Create the operation sequence (program, etc.). (3) Transfer and check signal and data with the external equipment. (4) Check operation at low speed (250 mm/sec or less). (2) During operation - Danger prevention measures that prevent contact with the robot (installation of the fence or enclosure) are taken to keep a person away from the operating robot. - When the operating robot is approached, measures shall be taken to turn off the servo power. (1) Start the robot. (2) Perform operation along with the purpose at the specified speed. (3) Stop the robot. (3) During maintenance and inspection - Contents performed within the movable range of the robot when the primary power is off (1) Check the bolts for looseness and backlash and re-tighten them. (2) Check the cable clamp for looseness. (3) Inspect damage. (4) Clean dust. (5) Inspect and apply rust prevention agent. (6) Check the timing belt for looseness and crack. (7) Check the cable and air tube for wear. (8) Check the operation when each axis is pushed and moved by hand. (9) Replace the consumable parts for the robot and controller. (10) Replace the battery for memory in the controller. - Contents performed within the movable range of the robot when the primary power is on (1) Check the grease state and apply grease (when the servo power is off). (2) Push each axis by hand and check it for backlash (when the servo power is on). (3) Check abnormal vibration and sound during operation (when the servo power is on). (4) Check the teaching point for misalignment (when the servo power is on). (5) Replace the battery for robot position detector (when the servo power is off).

Table 2　Restricted Specification Sheet (2)

2) Predictable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot body, controller, and optional parts and energization with these covers removed - Operation in states other than the specified installation state (placed on floor, suspended from ceiling, mounted to wall, etc.) - Operation on the frame that does not satisfy the specified strength - Operation in other than the specified environment
3) Unexpected start	- Malfunction caused by excessive noise - Robot start caused by start signal unexpectedly sent from the peripheral equipment to the robot - Unintended start caused by abnormal communication data - Malfunction caused by voltage fluctuation

2. Replacement of product components (restrictions on time)		
1) Mechanical restriction 2) Electrical restriction	Replace consumable parts as required or periodically through daily inspection, regular inspection (every 3 and 6 months and every year) and overhaul (every 5 years).	
3 Movable range of product (restriction on space)		
1) Operating range	Each axis	Check the specifications manual for your robot.
	Additional axis	Compliant with the individual specifications.
	—	—
	—	—
2) Interface	- The product shall be installed at a location suitable for outer shape and operating range. - Space required for maintenance and inspection shall be secured.	
3) Work environment	- Danger prevention measures that prevent contact with the robot (installation of the fence or enclosure) shall be taken. '- Space shall be secured to prevent a person from being caught between the robot and the fence or enclosure.	
4. Life cycle of product		
During the life cycle of assembly, shipment test, transportation, installation, teaching, operation, maintenance and disposal, risk assessment is carried out for “installation” stage (transportation and installation), “use” stage (teaching, operation, and maintenance) and disposal.		
5. Person who approaches the machine		
Target	Knowledge, experience, and condition	
1) Transportation and install	Knowledge: Person who has understood the instruction manual	
	Experience:—	
	Condition: Qualified person on forklift or crane (as required)	
2) Teaching/adjustment operation worker (including co-worker)	Knowledge: Person who has understood the instruction manual	
	Experience: Person who has acquired the operation of the robot to be used	
	Condition: Person who has received special training pertaining to the operation such as teaching of the industrial robot and to whom work has been permitted by the employer	
3) Operation worker (operator)	Knowledge: Person who has understood the instruction manual	
	Experience: Person who has understood the work regulation	
	Condition:—	
4) Maintenance and inspection worker	Knowledge: Person who has understood the instruction manual	
	Experience: Person who has acquired the operation of the robot to be used and person who has acquired maintenance work of the target robot	
	Condition: Person who has received special training pertaining to the operation such as inspection of the industrial robot and to whom work has been permitted by the employer	
5) Disposal worker	Knowledge: Person who has understood the instruction manual and the disposal method for each local government	
	Experience:—	
	Condition:—	
6. Other		

Warranty

We conduct a strict tests and inspections before delivery to ensure that the product satisfies our performance standards.
We provide warranty to cover defects of the product based on the following prerequisites.

1. Warranty period — The warranty period of the product is either of the following, whichever is earlier.
   (1) 18 months after delivery from our factory
   (2) 12 months after installation at your site

2. Warranty scope
   (1) We provide warranty for the product. Note that only the specifications and functions defined in the product specifications, catalog, and instruction manual are covered by the warranty. Any secondary or incidental damages caused by failure of the product are not covered by the warranty in any circumstances.
   (2) We provide repair of a faulty product for free only if the product has been handled and used according to the instruction manual that comes with the product.
   Any repairing after the warranty period passes will be charged.

3. Disclaimer — The warranty shall not apply to the following cases.
   (1) Any failure or damage arising from your careless use or misuse that is not defined in the instruction manual.
   (2) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables[*1)]).
   (3) Any failure regarding sensuous phenomena such as noise not affecting the functions
   (4) When any modification or disassembly without our consent is confirmed
   (5) Any failure or damage arising from insufficient maintenance and inspection or inappropriate repair
   (6) Any failure or damage due to act of God, fires, and other force majeure
   (7) Internal data such as programs and points created or modified by users
   (8) When a product purchased in Japan is taken abroad

1) Consumables are listed in 7 Replacement Parts for Maintenance.

4. Precautions
   (1) SHIBAURA MACHINE shall not guarantee the standard performance of the product if you use the product beyond the specifications.
   (2) SHIBAURA MACHINE shall not be liable for any injury, equipment damage or failure resulting from failure to observe the warnings and cautions stated in the instruction manual.
   (3) Note that warnings, cautions, and other information are those assumed by SHIBAURA MACHINE.

Standards, laws and regulations regarding safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" defines measures for reducing risks regarding machinery.

Step 1: Fundamental safe design measures - Restrictions on force, speed, and energy

Step 2: Safety protection - Installation of protective fences, etc.
Additional protection measures - Installation of emergency stop devices

Step 3: Information for use - Warning labels, alerts, instruction manuals

Based on the above, the ISO/IEC defines the "Guide 51" that groups various standards in a hierarchal structure. Safety standards for industrial robots are standardized as separate safety standards for machinery in layer C.

- ISO 10218, ISO 10218-1

Comply with the standards, laws and regulations in the countries where you use industrial robots.

Safety Precautions

The instruction manual for the controller contains important safety information for ensuring the safe and correct operation of this device and for preventing hazards to the operators and others and property damage.
To ensure that the procedures from installation to operation of the controller are performed safely, be sure to carefully read the separate instruction manual STEA1179_THE Series Safety Manual before starting actual operation.
Read and understand the meanings of the following indications and symbols first, and observe these precautions.

[Meanings of Indications]

Indication	Meaning
⚠ DANGER	Indicates the imminent danger that incorrect operation could cause death or serious injury.
⚠ WARNING	Indicates the danger that incorrect operation could cause death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury to[1)] to the operator.
NOTICE	Indicates the assumed possibility of only damage to property[2)].

*1: Injury indicates injuries such as wounds, burns, and electric shock that do not require hospitalization or long-term outpatient treatment.
*2: Damage to property indicates extensive damage related to the destruction of assets or materials.

⚠ DANGER
In order to ensure controller safety from transportation to disposal, be sure to carefully read the safety manual shown in STEA1179_THE Series Safety Manual. Performing an operation different from what is described in the manual could result in harm to people.

Service Network

Please refer to our distributor's handouts.

Specifications

This part explains the basic specifications and the names of components of the TS5000.

1. Specifications

1.1 Equipment Configuration

The TS5000 supports the THE SCARA robot.

For details how to connect the 200 V power supply, refer to 4.1.2 "ACIN" connector for connecting the power cable (Figure 4.1 - ①). Also, for details on how to connect the 24 V power supply, refer to 4.1.4 "SAFE" connector for connecting input/output external power supply and safety input/output (Figure 4.1-⑥).

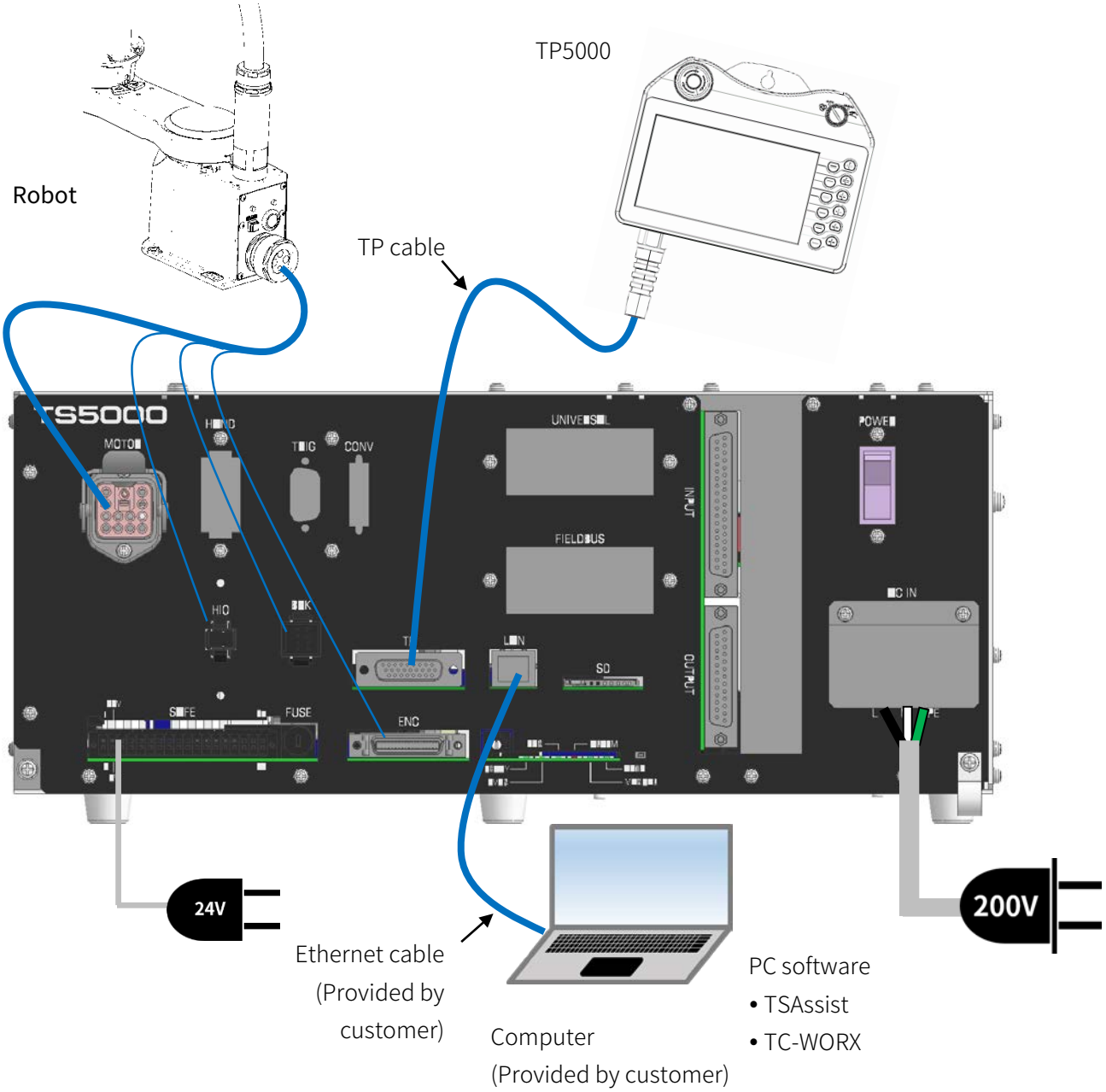


Figure 1.1 Equipment configuration

1.2 Names of Components

The connections for the power supply terminal, motor control cable, and SAFE connector are explained in "4.Cable Connection".

Options

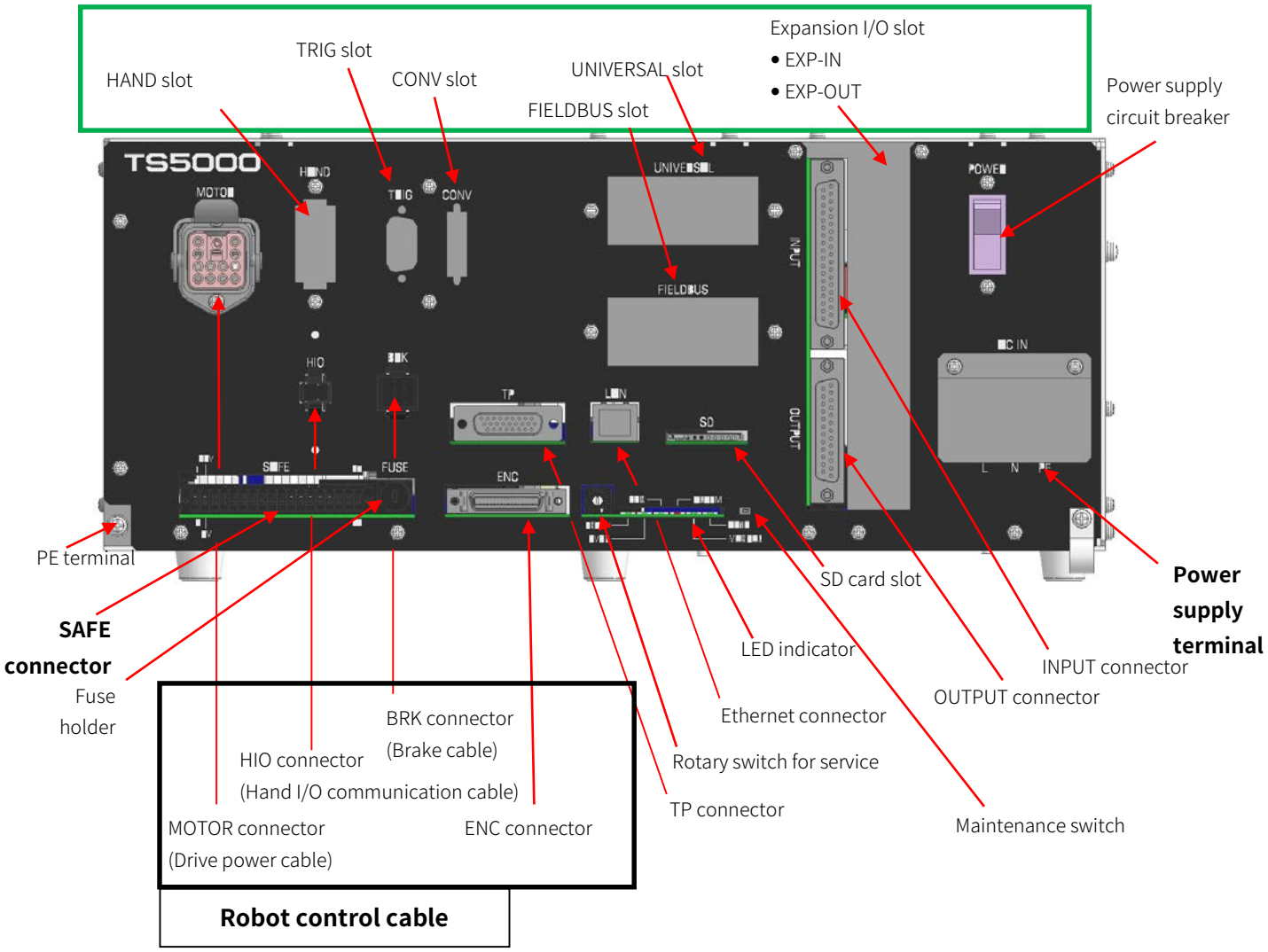


Figure 1.2 Names of components (front view)

MEMO

1. *The rotary switch for service is used by SHIBAURA MACHINE service department. The memory is cleared, and so never touch the rotary switch.*

1.3 Controller Specifications

No.	Item		Specifications		Remarks
1	Number of controlled axes		4 axes		
2	Servo type		Digital servo		
3	Position detection method		Absolute type		
4	Main memory		Controller built-in Flash ROM Capacity: 12 MB		Write count limit 10,000
5	Auxiliary memory		SD card (SD, SDHC) Maximum capacity: 32 GB		SDXC is not supported
6	Number of program stores		Main memory	Up to 512 User files: 502 System files: 10	
			Auxiliary memory	Up to 512 User files: 512	
7	Maximum number of program lines		Per program Data section 5000 points Program section 5000 lines		
8	Programming language		SCOL2		
9	Operation command		PTP, CP (line, arc), shortcut, arch		
10	Teaching device		Teach pendant, program support software		Option
11	Teaching method	Point teaching	Guidance operation from teaching device		
			Coordinates: Input the X, Y, Z, and C values via the teach pendant.		
			Servo-free: Arm moves by input		
		Program input	Input from teaching device		
12	Master modes		MANUAL mode/AUTO mode		
13	Execution modes		Continuous, cycle, step, step for each operation		
14	Speed setting		Override/Speed limit/Program instructions, 1 to 100% for each instruction		
15	Acceleration setting		Program instruction: 1 to 100%		
16	Torque Control		Program instruction: 1 to 300%		
17	Coordinate system		The Base, Workpiece, and Tool coordinates can be set.		
18	Movement limit		Soft limit		
19	Self-diagnostic function		Alarm detection, etc.		
20	Interrupt function		Starts an interrupt program via an input signal, timer, etc.		
21	Communication function		Ethernet 1 ch (1 Gbps)		

22	I/O	General-purpose	8 inputs/8 outputs	
		System	13 inputs Program selection, start, stop, reset, etc. 9 outputs Servo ON, operation preparation complete, fault, etc.	
		Hand	8 inputs/8 outputs	
		Extension	21 inputs/17 outputs	Option
23	Dimensions		365 (W) × 161 (H) × 350 (D)	The height includes the rubber foot height.
24	Weight		11 kg	
25	Power supply		Single phase 190 to 240 VAC, 50/60 Hz	
26	Ambient operating temperature		0°C to 40°C	
27	Ambient storage temperature		-10°C to 50°C	
28	Lifespan parts		Button cell battery (replacement recommended after 5 years) Switching power supply (replacement recommended after 5 years) Large-capacity capacitor (replacement recommended after 9 years)	The actual lifespan will vary depending on the usage conditions.

MEMO

1. *The compatible SD cards are SD and SDHC.*
2. *Use an SD card formatted with FAT16 or FAT32.*

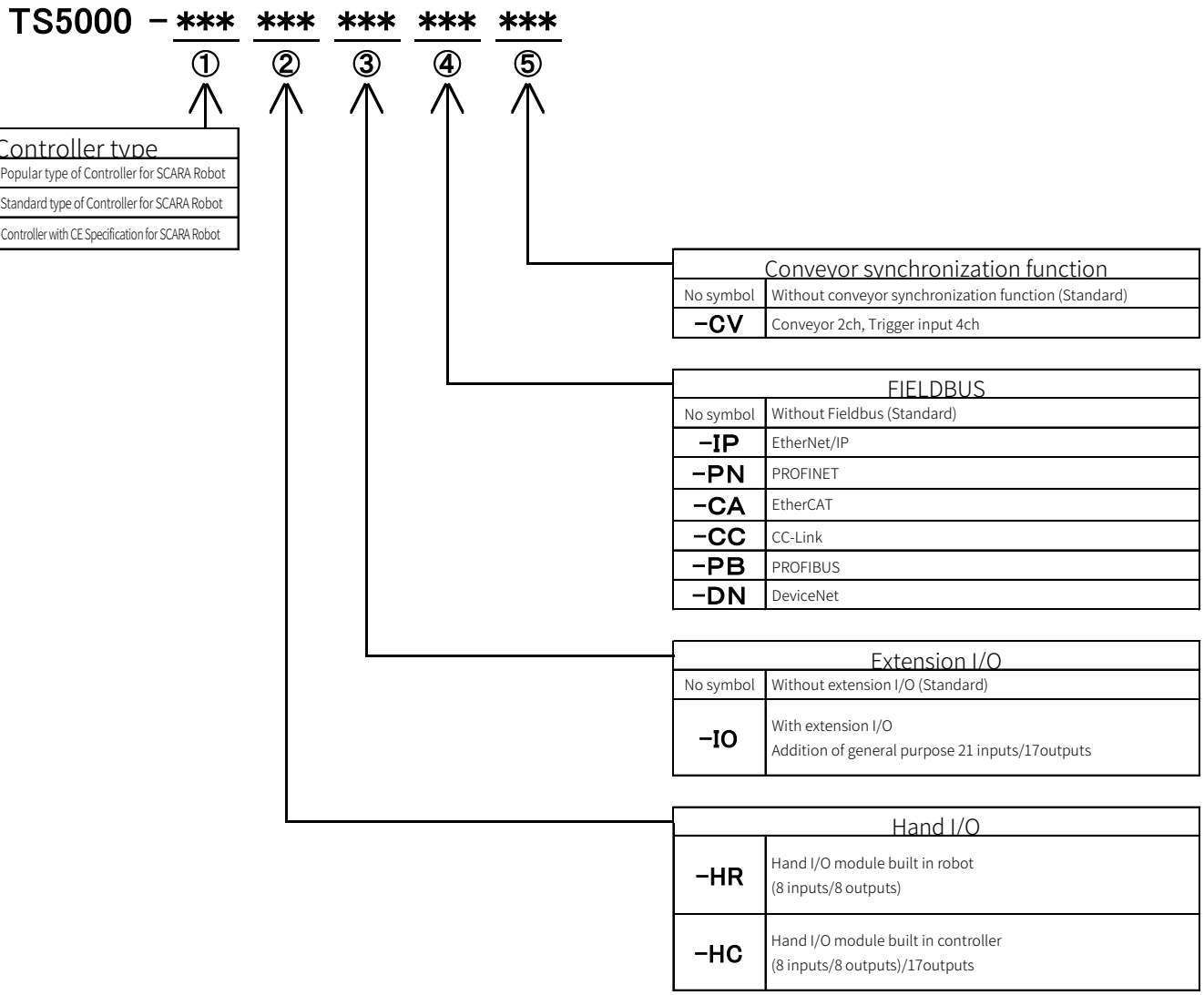
1.4 Specifications of Controller Options

No.	Name of product		Remarks
1	Teach pendant	TP5000	Standard cable 5 m
2		TP1000	To connect the TP1000, the "TS5000-TP1000 conversion cable" is required. 1)
3	TS5000-TP1000 conversion cable		
4	Controller mounting bracket		
5	Extension input/output signal		21 inputs/17 outputs
6	Program support software	TSAssist	Windows application
7	TCmini (built-in PLC) editing software	TC-WORX	Windows application
9	Fieldbus function	CC-Link	Under development 2)
10		EtherNet/IP	
11		EtherCAT	
12		PROFINET	
13	Conveyor synchronization function		
14	Vision + Conveyor synchronization function		

1): Teach Pendant TP3000 cannot connect to the TS5000.

2): The Fieldbus slave module type can be selected from CC-Link, EtherNet/IP, EtherCAT, or PROFINET. The Fieldbus to be used must be determined when the order is placed.
CC-Link is a registered trademark of Mitsubishi Electric Corporation. EtherNet/IP is a registered trademark of ODVA, Inc. EtherCAT is a registered trademark of Beckhoff Automation GmbH & Co. KG. PROFINET is a registered trademark of PROFIBUS Nutzerorganisation e.V.

1.5 Controller Model Name



Controller type	
SS	Popular type of Controller for SCARA Robot
MS	Standard type of Controller for SCARA Robot
EMS	Controller with CE Specification for SCARA Robot

Conveyor synchronization function	
No symbol	Without conveyor synchronization function (Standard)
–CV	Conveyor 2ch, Trigger input 4ch

FIELDBUS	
No symbol	Without Fieldbus (Standard)
–IP	EtherNet/IP
–PN	PROFINET
–CA	EtherCAT
–CC	CC-Link
–PB	PROFIBUS
–DN	DeviceNet

Extension I/O	
No symbol	Without extension I/O (Standard)
–IO	With extension I/O Addition of general purpose 21 inputs/17outputs

Hand I/O	
–HR	Hand I/O module built in robot (8 inputs/8 outputs)
–HC	Hand I/O module built in controller (8 inputs/8 outputs)/17outputs

Installation

This part explains transport and installation of the controller.

2. Transportation, Unpacking, and Carrying

Transportation, unpacking, and carrying of the controller must be performed by qualified personnel. Also, these processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

2.1 Controller Transportation

⚠ CAUTION

- Do not stack the packaging boxes on top of each other. The load could topple over, resulting in an injury.
- If the controller is carried manually, be sure to wear safety gloves and safety shoes. Your fingers, hands, or feet could get trapped by the controller.
- Use a cart or similar device when carrying. The controller could be dropped if carried by hand.

NOTICE

- Do not subject to excessive vibrations during transportation. This could damage the controller.
- Do not transport the controller upside-down or store it in a vertical orientation. This could damage the controller.
- Do not allow moisture to get inside the controller. This could cause a breakdown.

2.2 Unpacking the Controller

The controller is shipped in corrugated cardboards. Open the packages in a location easily assessable, where the equipment is to be installed. Take careful precautions not to damage the robot and controller. After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport. For the controller accessories, see 8.3 List of Accessories.
After unpacking, dispose of any unneeded wooden pallets, cardboard boxes, plastic bags, cushioning materials, and other items in compliance with the regulations of the customer.
Perform the same when repacking.

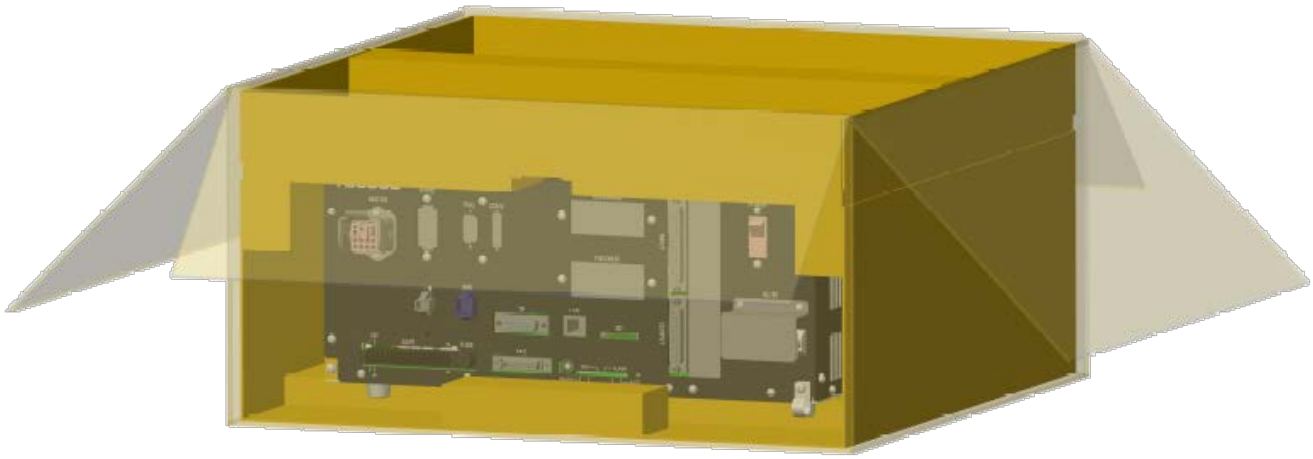


Figure 2.1 Controller packing state

⚠ CAUTION

- Wear protective gear when disposing of the unneeded pallets and cardboard boxes after unpacking. Otherwise, you could be cut by nails sticking out and by other sharp locations, resulting in an injury.
- Use a dedicated cardboard cutter when opening the packing box. You could cut your finger or hand if using a cutter with a sharp edge.
- Unpack at a stable location. If unpacking at an unstable location is performed, the controller could fall, and you could get your hands or feet trapped.
- Wear safety gloves when unpacking. Your hand could get trapped between a cardboard box and the controller, resulting in an injury. You could also be injured by the corners of the controller or cardboard box.

NOTICE

- Do not allow liquid such as water or foreign material to get inside the controller. This could cause a breakdown.

2.3 Transport of the Controller

⚠ CAUTION

- Do not carry by holding the AC power supply unit. If the controller is carried by holding the AC power supply unit, it could be damaged and fall, and your hands or feet could get caught. Also, an electric shock could occur if AC power is being supplied.
- If the controller is transported manually, be sure to wear safety gloves and safety shoes. Your fingers or feet could get caught.
- Use a cart or similar device when carrying and transporting. If the controller is transported by hand, your fingers or feet could get caught.
- Transport the controller while the controller cables are removed. If the controller is transported while the cables are still connected, a cable could be broken, resulting in an electric shock.

NOTICE

- Do not subject to excessive vibrations. This could damage the controller.
- Do not allow liquid such as water or foreign material to get inside the controller. This could cause a breakdown.

2.3.1 Precautions for storing the controller

NOTICE

- Store the controller avoiding direct sunlight.
  Otherwise, the controller could become hot inside, resulting in a breakdown.
- Seal the controller in a vinyl bag with desiccant inside to prevent rust and dust.

3. Installation

The installation of the controller can be carried out only by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

⚠DANGER

- Never install the controller in a location where the water splashes on it. Otherwise, water could get inside the controller to cause the circuit to be short-circuited, resulting in a fire or an electric shock due to ground leakage.
- Never put near flammable objects. If sparks occur due to a breakdown, nearby flammable objects could ignite, resulting in a fire.
- Do not place objects on top of the controller. (Stacking prohibited)
  - If there is not adequate space for ventilation, heat could build up inside, resulting in a fire.
  - The controller is designed assuming that objects would not be placed on top of it, and so the top could be unable to withstand the weight, causing your hands or feet to get caught.
- Be sure to always install the controller outside the hazard zone and at a location where robot operations are visible.
  If the controller is operated within the hazard zone or the controller is used at a location where the robot or peripheral devices are not visible, the operator or worker within the hazard zone could collide with the robot or peripheral device.
- If it is unavoidable to install the controller in the hazard zone, take the teach pendant out of the hazard zone and use the system signal of the controller and output the robot status externally to a revolving light or LED for indication so that the operator can check the robot status through them.

If there is no means to check the robot status, the operator who unavoidably tries to check the controller installed in the hazard zone could collide with the robot or peripheral device.

- If the controller is mounted in an airtight space such as in a control panel, provide air holes, forced ventilation, and indirect cooling so that no heat buildup occurs. If a controller is mounted in an airtight space, heat could build up, resulting in a fire.
- Either mount the controller inside the control panel so that it cannot be contacted by workers, or mount in a location that can provide sufficient distance from workers. If the controller is installed within the movable range of workers, they could get their hands or feet entangled in the cables, causing them to trip and fall, or receive an electric shock due to contact with current-carrying parts.
- Do not install the controller on an unstable location. The controller could fall, and your hands or feet could get caught.
- Connect the robot wiring after installation is complete.
  - If the controller is installed or transported while the wiring is connected, a cable could be broken, resulting in an electric shock.
  - Feet could get entangled in the wiring, resulting in a fall.
- Use a dedicated duct to protect the lead-in cables of the control panel. If the cables are pulled around without using the dedicated duct, the cables could be broken, resulting in a fire or an electric shock could be caused due to ground leakage
- Mount a cover or other enclosure on the buttons and switches on the control device connected to the controller. If a signal is turned on due to unintentional contact, the robot could be activated, resulting in a collision.
- Display the operation mode of each controller on the operation device. If you are unaware of the operation mode, a collision could occur with a robot that is performing an unexpected operation.

NOTICE

- Never supply a current that exceeds the rated output. If a current exceeding the rated output is supplied, the output elements, boards, or other components could be damaged.
- Do not place objects on top of the controller. (Stacking prohibited)
  - The controller is designed assuming that objects would not be placed on top of it, and so the top could be unable to withstand the weight and collapsed, damaging the controller.
- If the LAN, OUTPUT, or SD connectors will not be used, be sure to always mount a connector cap (a dummy SD card for the SD connector). If no protection is provided, a breakdown could occur due to an electrostatic charge.
- If the controller is mounted in an airtight space such as in a control panel, provide air holes, forced ventilation, and indirect cooling so that no heat buildup occurs. If a controller is mounted in an airtight space, heat could build up, resulting in a breakdown.

3.1 Installation Environment

Table 3.1 shows the environment specifications for the controller.

Table 3.1 Environment specifications for the controller

Item	Specifications
Temperature	In operation state: 0 to 40°C In storage state: -10°C to 50°C
Humidity	20 to 80% (no condensation) Do not install the controller in a place where liquid such as water is applied.
Altitude	Up to 1000 m
Vibration	Do not apply excessive vibration and shock.
Dust	Ensure that there is no electric conductive dust. Contact us if you use the controller in a dusty environment.
Gas	Ensure that there is no corrosive or flammable gas.
Protection degree	IP20
Overvoltage category	IEC 60664-1 Class III
Electric shock protection	IEC 61140 Class I
Pollution level	IEC 60664-1 Pollution Level 3
Solar radiation	Avoid direct sunlight.
Power supply noise	Ensure that there is no surrounding device generating excessive noise.
Magnetic field	Ensure that there is no surrounding device generating a strong magnetic field.
Surrounding environment	Ensure that there is no iron powder, oil, salt, or organic solvent in the environment. Ensure that water is not applied to the controller.

3.2 Dimensions

Figure 3.1 shows the outline drawing of the controller. The weight of the controller body is 11 kg.

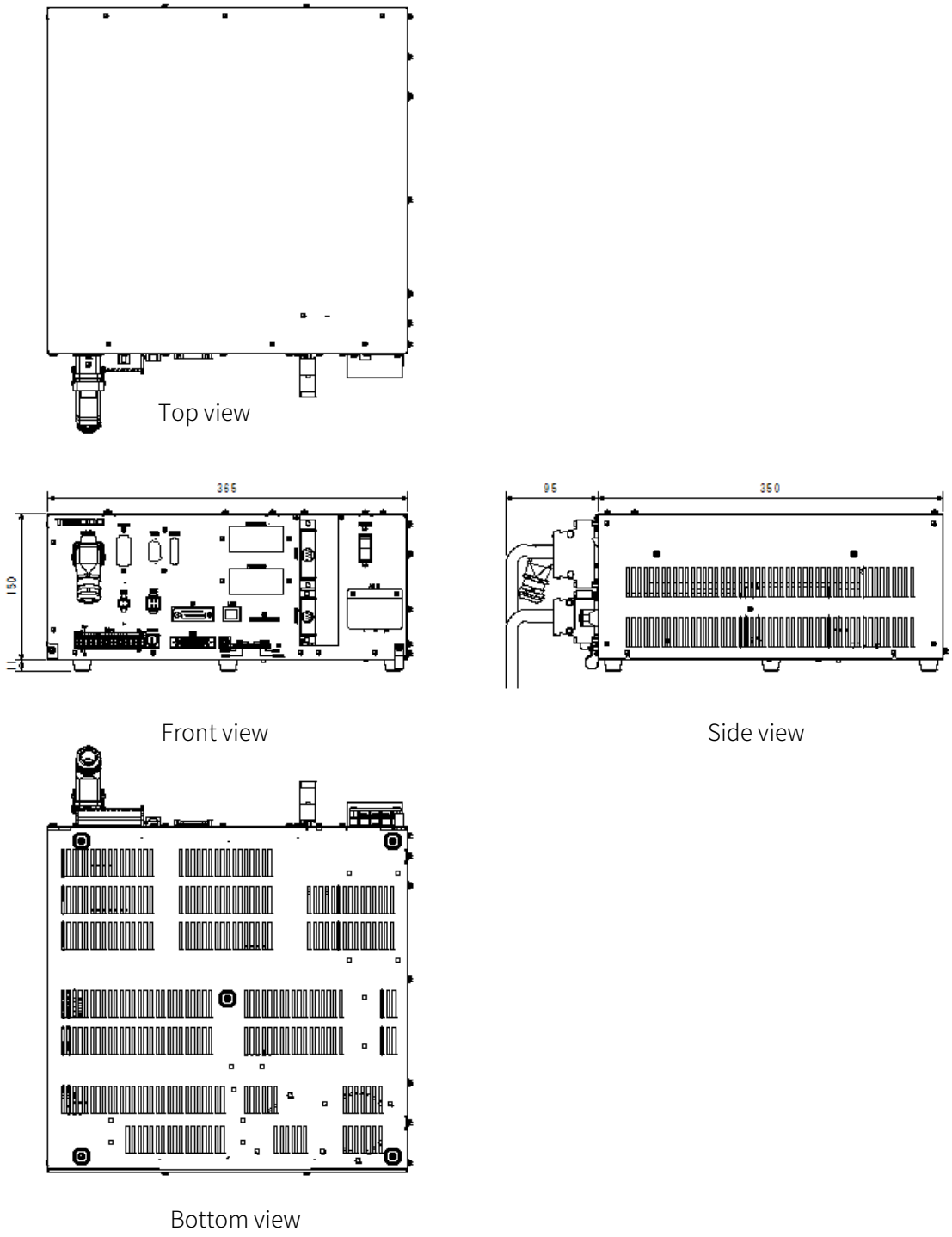


Figure 3.1 External views

3.3 Precautions for Installation on Floor

Figure 3.2 shows the ventilation space for the controller.

⚠ DANGER

- If the controller is mounted in an airtight space such as in a control panel, provide air holes, forced ventilation, and indirect cooling so that no heat buildup occurs. If the controller is mounted in an airtight space, heat could build up, resulting in a fire.

NOTICE

- To ensure adequate ventilation around the controller, be sure to provide at least 50 mm of clearance on the right, left, and rear sides, at least 100 mm of clearance on the top side, and a clearance equivalent to at least the height of the rubber feet (11 mm) on the bottom side. If the ventilation space is not provided, the controller could overheat and break down.
- When mounting the controller in the control panel, use anchoring fixtures and other implements to secure the controller in place. Otherwise, the controller is subjected to vibration, resulting in a breakdown.

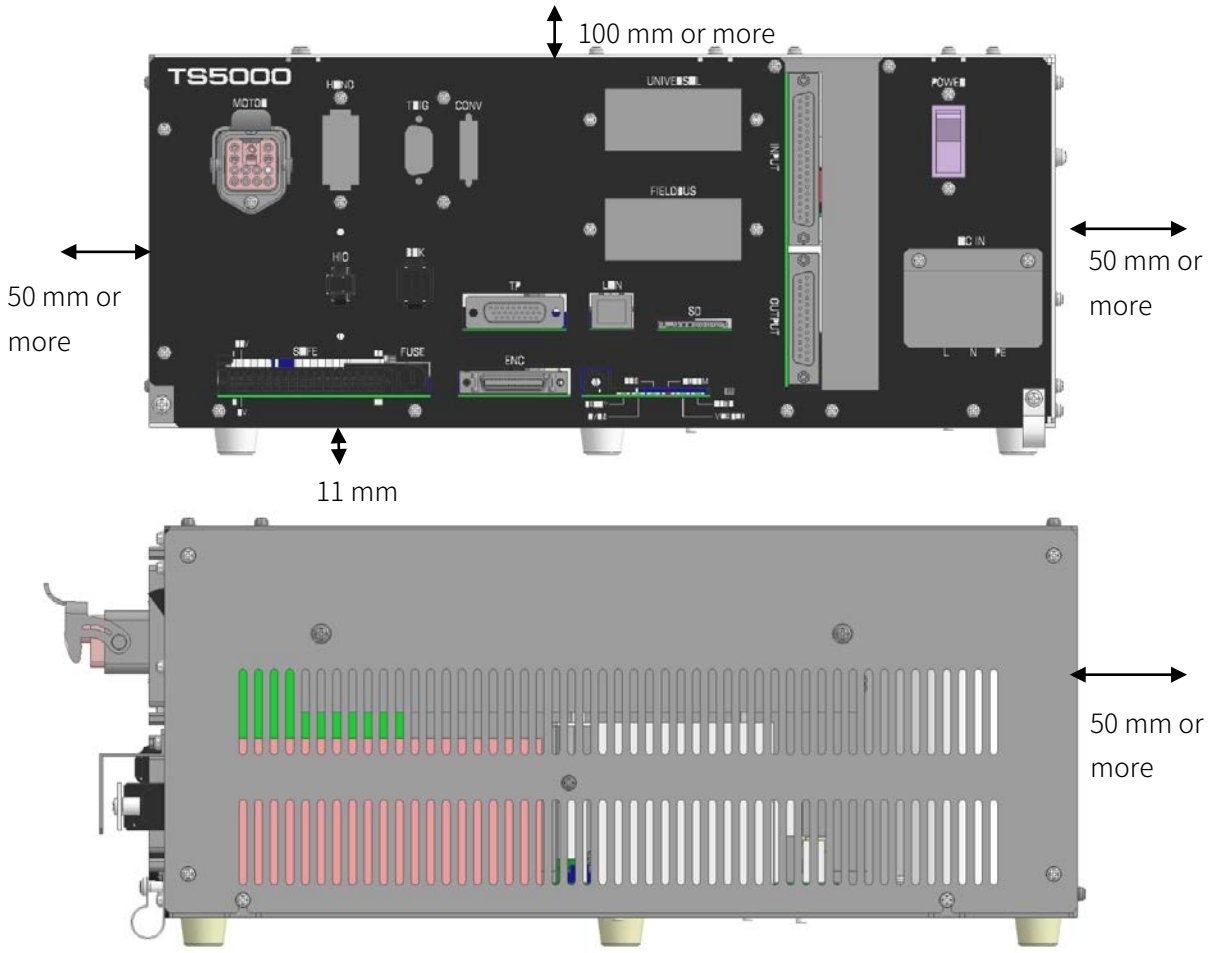
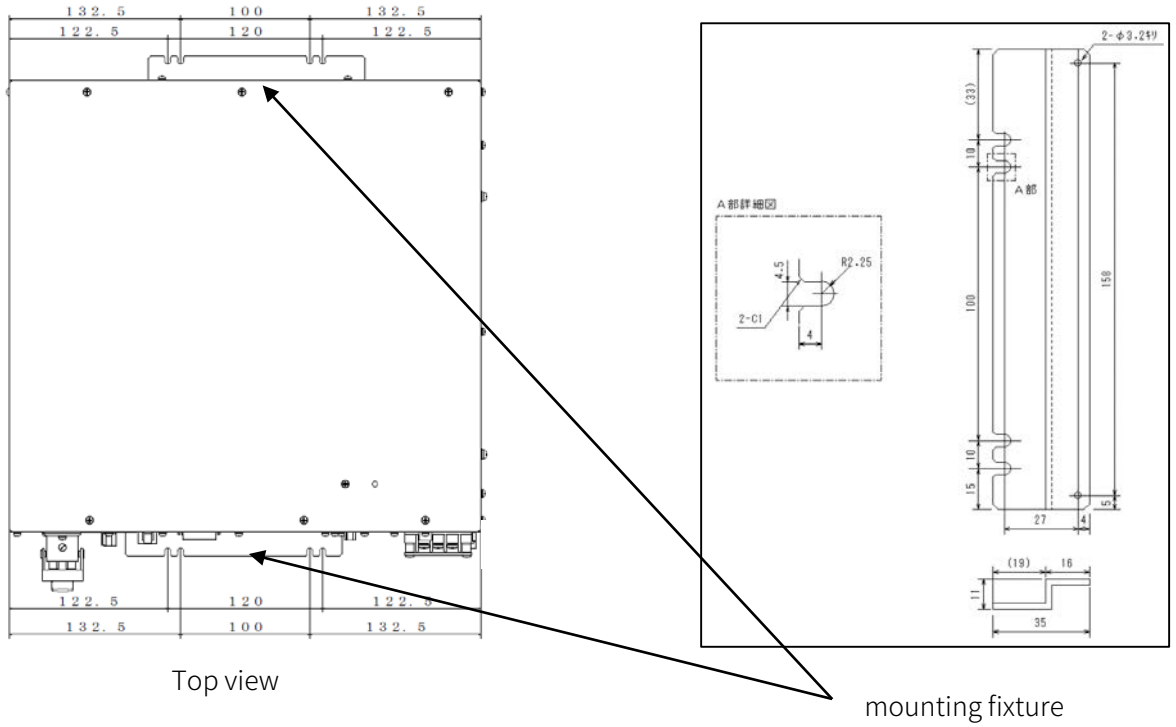


Figure 3.2 Ventilation space for the controller

3.4 Mounting Surface Dimensions

If the controller will be mounted to the control panel or other location, use a dedicated mounting fixture and secure the controller to the control panel with M4 screws. Mounting is performed by the customer. For consultation concerning the mounting fixture (option), contact the Service department or Headquarter Sales department.



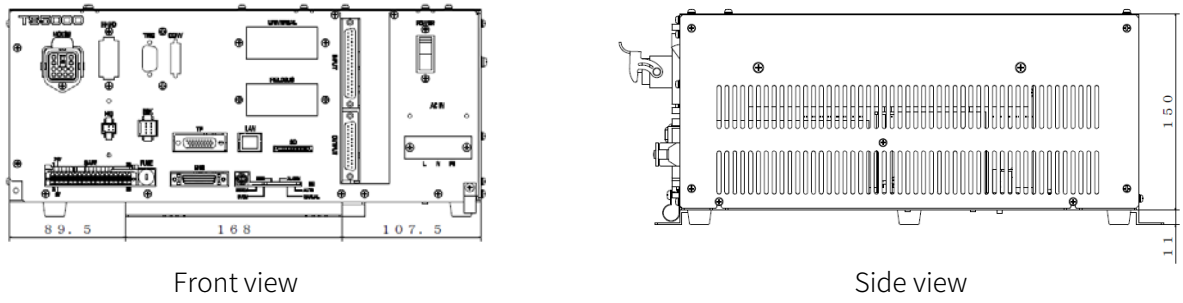


Figure 3.3 Dimensions of the controller's fixation parts

NOTICE

- When mounting the controller in the control panel, use anchoring fixtures and other implements to secure the controller in place. Otherwise, the controller is subjected to vibration, resulting in a breakdown.

3.5 Notes when mounting the controller to the control panel or other sections

Observe the following when mounting the controller to the control panel, etc.

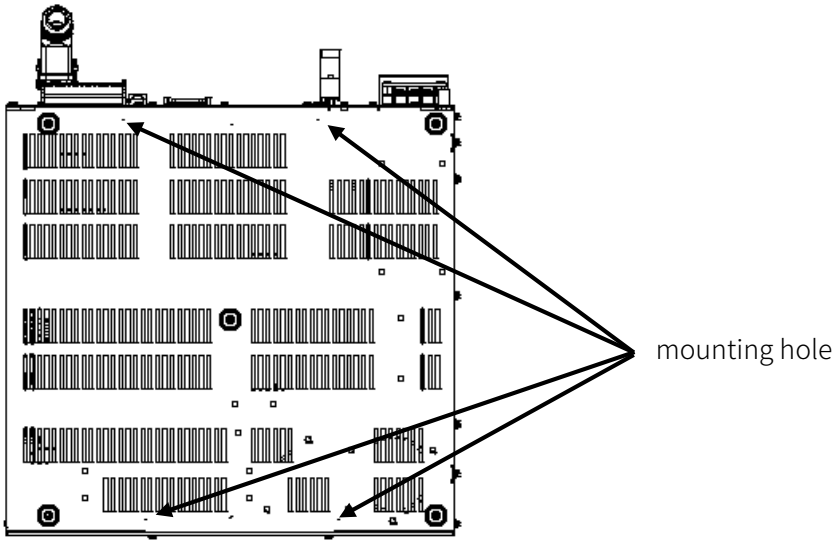
(a) When installing the controller on the control panel, remove the rubber feet from the bottom surface, and attach the mounting brackets to the mounting holes.

(b) When mounting the controller, set the controller in a location where there are no impediments to performing maintenance of the controller. In particular, when the controller is housed in the control panel, it must be taken out from the control panel when maintenance is performed.

(c) Arrange so that the cables connected to the controller can be pulled out.

(d) If taking out the controller from the control panel to perform maintenance, use a configuration where all cables can be reconnected, and the robot can be operated.

(e) Provide a 95mm or more clearance to enable the connection of the robot's cable connectors to the controller's front section.



Bottom view

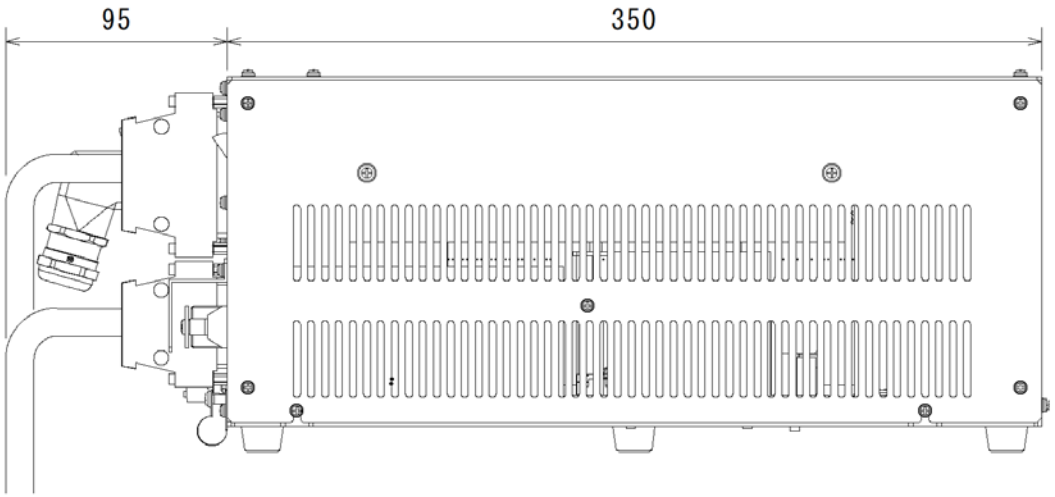


Figure 3.4 Clearance from the front of the controller

4. Cable Connection

Cable connection can only be carried out by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

⚠DANGER

- Never use unauthorized cables. Manufacturing and use of unauthorized cables can result in a fire.
- Never connect to unauthorized connectors or terminals. Wiring of unauthorized connectors or terminals can lead to a fire.
- Always connect the ground wire. Usage without connecting the ground wire can result in ground leakage, electric shock, or a fire.
- After connecting, use a tester or other instrument to check the voltage and for short-circuits. When there is miswiring or during controller breakdown, turning on the power could result in a fire or an electric shock due to ground leakage.
- Firmly connect the connectors. If the connectors do not fit firmly, electrostatic charges could build up and lead to a malfunction, which could result in a collision with a robot.

NOTICE

- Do not connect to unauthorized connectors or terminals. Wiring of unauthorized connectors or terminals can lead to a breakdown of the controller.

4.1 Wiring

This section describes types of cables and connectors and how to connect them.

4.1.1 Connectors on the controller

Figure 4.1 shows the controller connector layout. The following sections explain the connections for cables ① to ⑦. For details on pin assignment of various connectors and other cable connections, refer to the STEA1167 Interface Manual.

① "ACIN" connector for connecting AC power cable
② "MOTOR" connector for connecting robot drive power cable
③ "HIO" connector for connecting hand I/O communication cable
④ "BRK" connector for connecting brake cable
⑤ "ENC" connector for connecting encoder cable
⑥ "SAFE" connector for connecting input/output external power supply and safety input/output
⑦ TP connector

Robot control cables

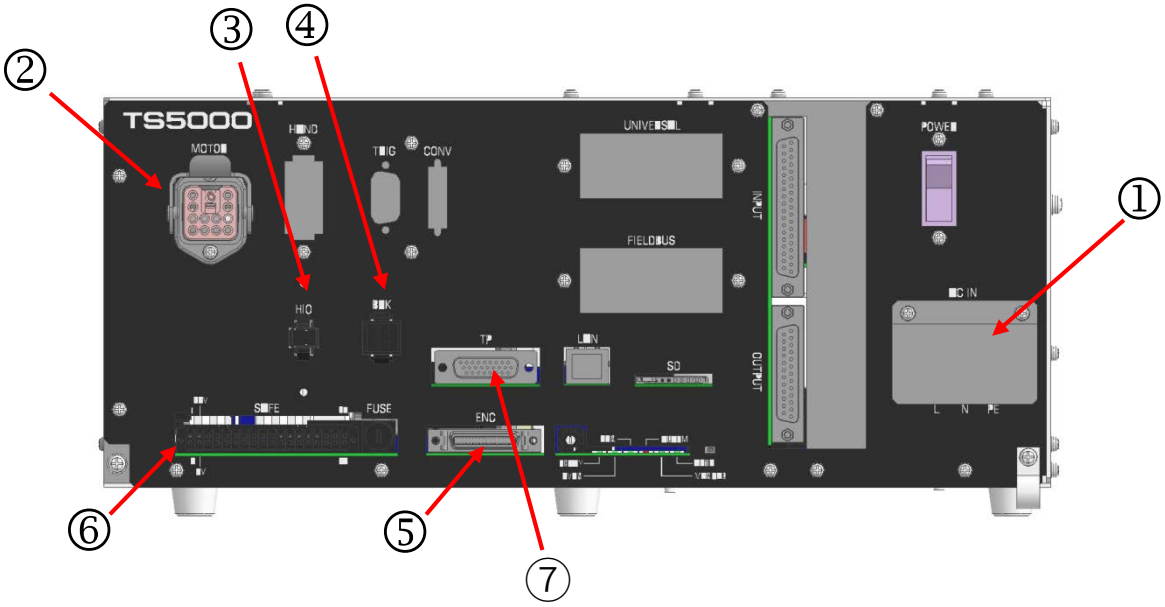


Figure 4.1 Controller connector layout

4.1.2 "ACIN" connector for connecting the power cable (Figure 4.1 - ①)

The power cable is used for supplying 200 VAC to the controller. The cable is not included in the accessories. It must be manufactured by the customer. The ACIN connector is used (Figure 4.1-①).

Also, if the main power is not supplied properly to the controller due to a missing phase, low voltage, or other abnormality, the error "8-027 Slow Charge Error" occurs when the servo is turned on. If the error occurs, verify whether the supply voltage at the controller power connector satisfies the input power supply specifications value for the controller and whether the supply voltage is stable. For details on “8-027 Slow Charge Error”, refer to the STEA1165_Alarm Manual.

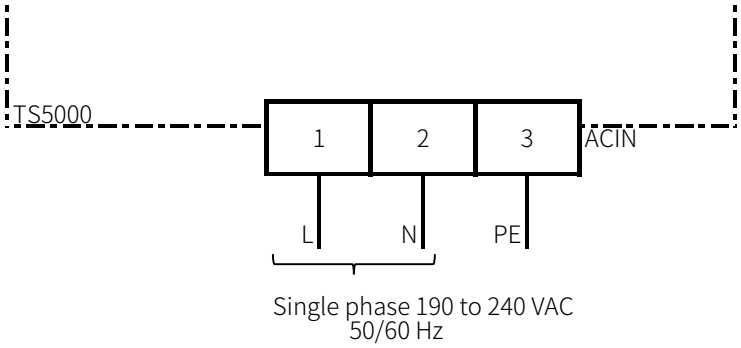


Figure 4.2 AC power cable wiring

Table 4.1 Power supply specifications

Power supply	Single-phase, 190 to 240 VAC 50/60 Hz ± 1 Hz
Momentary power loss	Within 40 msec
Grounding	Class D only
Connection screw size	Connect using M4 round terminal

⚠ DANGER

PROHIBITED

- Do not put conductive objects or fingers in the power supply terminal area. Otherwise, an electric shock could be received.

4.1.3 Motor control cable (Figure 4.1 - ② to ⑤)

The controller and robot are connected to the robot control cable “4.1.3.1 "MOTOR" connector for connecting robot drive power cable (Figure 4.1-②)”, “4.1.3.2 "HIO" connector for connecting hand I/O communication cable (Figure 4.1-③)”, 4.1.3.3 "BRK" connector for connecting brake cable (Figure 4.1-④), and “4.1.3.4 "ENC" connector for connecting encoder cable (Figure 4.1-⑤)”.

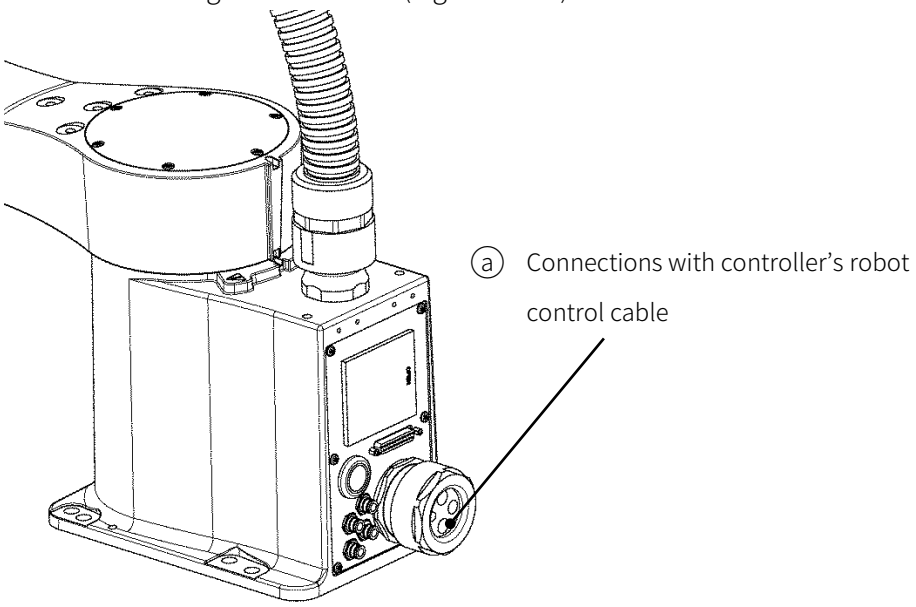


Figure 4.3 Robot-side connector layout

4.1.3.1 "MOTOR" connector for connecting robot drive power cable (Figure 4.1-②)

This cable connects the controller and robot, and it supplies the driving power needed for motor rotation to the various axis motors of the robot from the controller servo driver.

4.1.3.2 "HIO" connector for connecting hand I/O communication cable (Figure 4.1-③)

The hand I/O communication cable controls the hand on the robot tip, and in its standard configuration, is a signal cable capable of using 8 inputs and 8 outputs.

NOTICE

- Set the hand output signal to 0.1 A or less per output (total for 8 outputs is 0.8 A or less). If a current exceeding the rated output is supplied, the output elements, boards, or other components could be damaged.

4.1.3.3 "BRK" connector for connecting brake cable (Figure 4.1-④)

The brake cable turns on and off the brake for holding the motor axes.

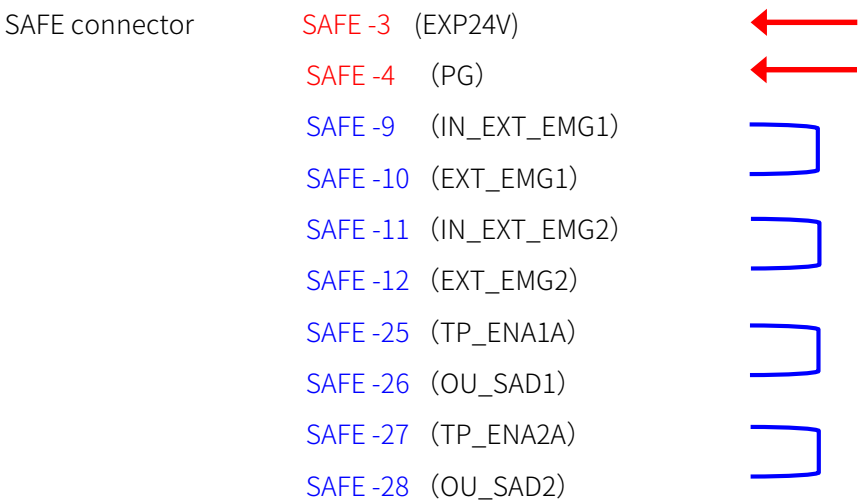
4.1.3.4 "ENC" connector for connecting encoder cable (Figure 4.1-⑤)

The encoder cable is a signal line to input signals from the rotation angle detection encoder of each axis of the robot to the controller.

4.1.4 "SAFE" connector for connecting input/output external power supply and safety input/output (Figure 4.1-⑥)

The SAFE connector supplies power for I/O and emergency stop signal input to the controller from an external device. The supplied connector B2CF 3.50/36/180LR is used for the emergency stop signal input section (safety input/output) configuration or for the supply of 24 VDC (I/O external power supply) from an external device. The I/O external power supply is used for system I/O, general-purpose I/O, extension I/O, and hand I/O. Because the connector does not have any printed pin numbers, connect the wires by aligning with the front silkscreen printing on the controller.

Also, if the safety input/output will not be used, the emergency stop line of this connector must be connected with a jumper to enable robot operation. To use the connector in the standard specifications, wire according to Figure 4.4. Four cables for the SAFE connector jumpers are included in the accessories.
To provide the emergency stop switch or safety door externally or connect the safety relay unit internally or externally, refer to “4.2 Connecting Input/Output External Power Supply Cable and Safety Input/Output Cable” in the STEA1167_Interface Manual.



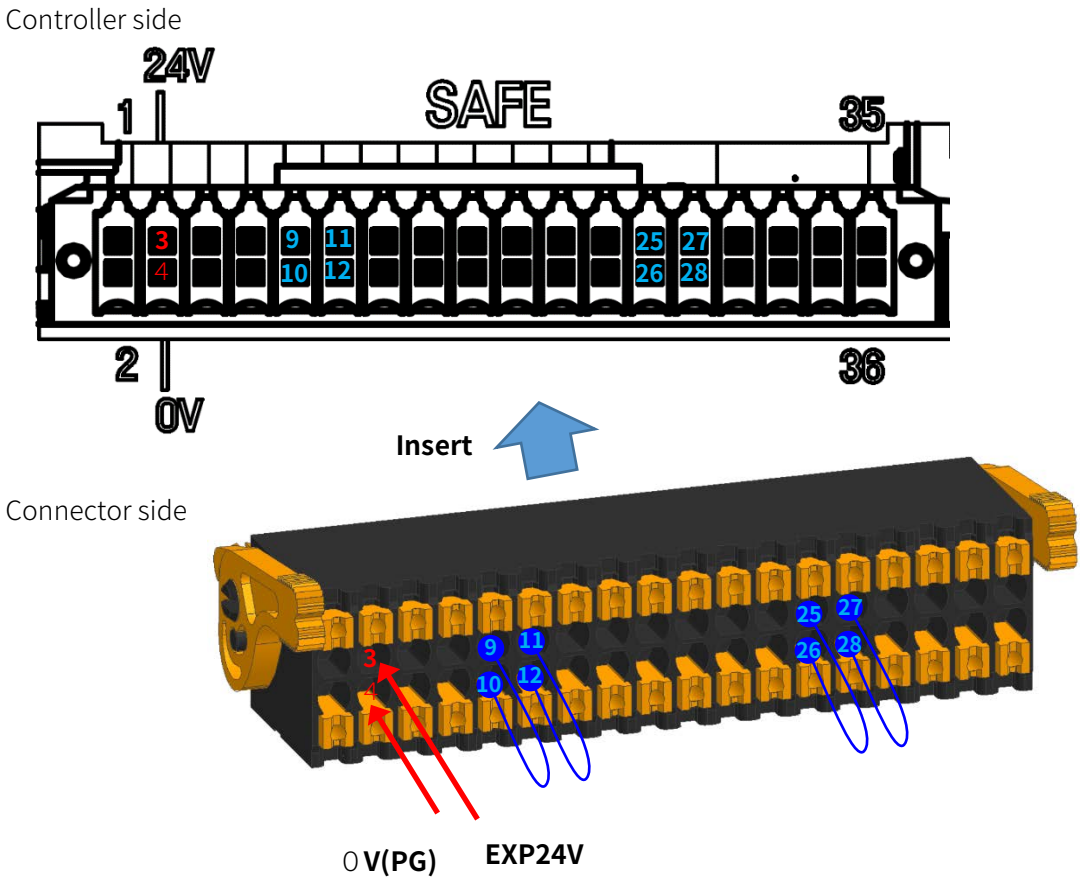


Figure 4.4 Jumper connection method for standard specifications (without safety relay module)

Table 4.2 I/O external power supply specifications

Power supply	24 VDC±10%
Supported cables	0.14 mm^2 to 1.5 mm^2 (AWG26 to AWG16)

NOTICE

- Apply a voltage in the range of 24 V±10% to the external input/output power cable. Applying a voltage that exceeds the allowable voltage could damage the controller.

4.1.5 Connecting/disconnecting the connectors

The TP connector is a connector used to connect a teach pendant to the controller. The optional TP5000 and TP1000 (requires a dedicated conversion cable) can be connected.
For details on the TP connector, refer to "7. TP Connector" in the STEA1167_Interface Manual.

MEMO
1. *If the TP connector is disconnected while in AUTO mode, an emergency stop will occur and Master mode will be maintained in AUTO mode.*

4.1.6 Connecting and disconnecting cables

⚠ WARNING

- Turn off the power before connecting and disconnecting cables from the controller. Touching the terminal for a cable that was removed while the power is on could result in an electric shock.
- When disconnecting cables, grasp the connector while holding the controller, and pull to remove.
  ・Pulling the cable itself could break the cable and result in an electric shock.

NOTICE

- Turn off the power before connecting and disconnecting cables from controller. Touching the terminal for a cable that was removed while the power is on could result in an electric shock, and connecting and disconnecting cables while the power is on could damage the ICs in the board.
- When disconnecting cables, grasp the connector while holding the controller, and pull to remove.
  ・Otherwise, the controller could topple over, resulting in a breakdown.

After the cable-side connector is connected to the controller, use the anchoring fixture to lock the connector connection in place. If the inserted connector is loose, this could lead to poor contact of the connector, resulting in an accident, and so check that it is firmly connected.
To remove a connector, remove the anchoring fixture, and then pull off the connector.

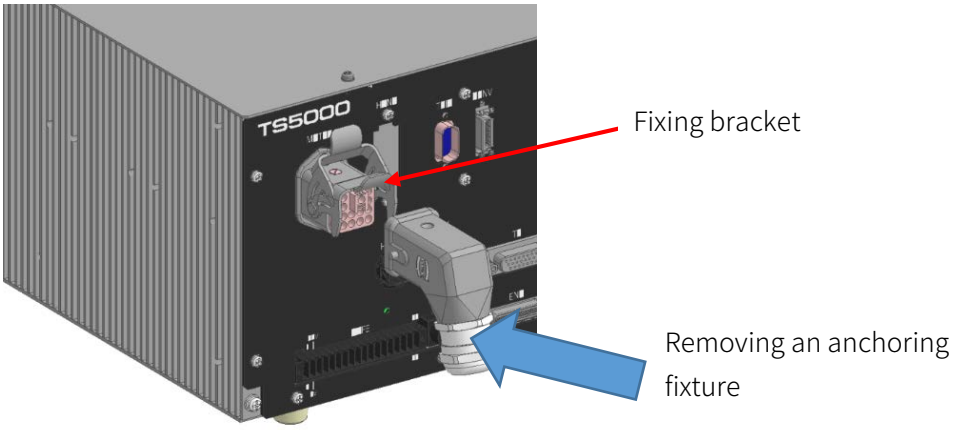


Figure 4.5 Attaching and removing the robot drive power cable

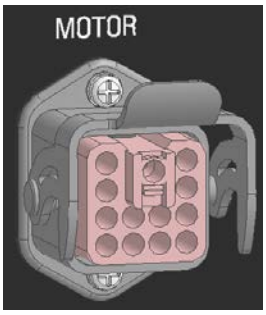
4.1.7 Connector terminal arrangement

4.1.7.1 AC power connector "ACIN" (Figure -①)

Controller side
Model: OGM-300NE-3P
Manufacturer: Osada Co., Ltd.

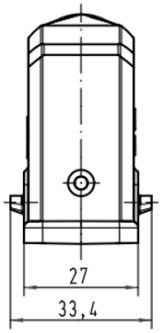


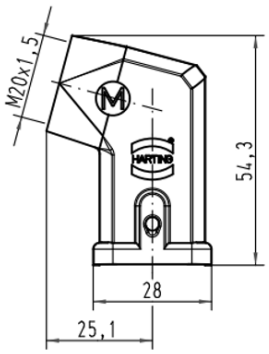
4.1.7.2 Robot drive power connector "MOTOR" (Figure -②)



Controller side
Model： 09120123101
09200030301
Manufacturer: Harting Co., Ltd.

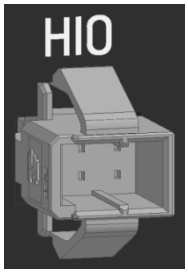
Cable side
Model： 19200031640
09120123001
Manufacturer: Harting Co., Ltd.





4.1.7.3 Hand I/O connector "HIO" (Figure -③)

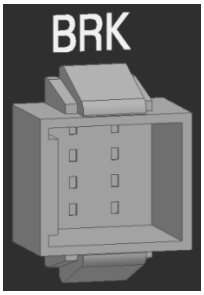
Controller side
Model: 1-1903131-2
Manufacturer: TE Connectivity Ltd.



Cable side
Model： 1-1827864-2
Manufacturer：TE Connectivity Ltd.
Maximum diameter：7.6×11.3 mm

4.1.7.4 Brake connector "BRK" (Figure - ④)

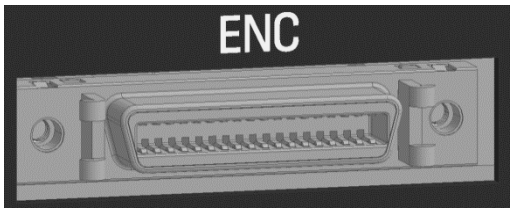
Controller side
Model: J21DPM-08V-KX
Manufacturer: J.S.T. Mfg. Co., Ltd.



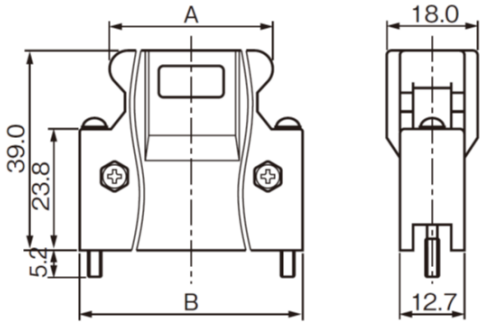
Cable side
Model: J21DF-08V-KX
Manufacturer: J.S.T. Mfg. Co., Ltd.
Maximum diameter：13.9×16 mm

4.1.7.5 Encoder cable connector "ENC" (Figure -⑤)

Controller side
Model: 10236-52A2PL
Manufacturer: 3M Company

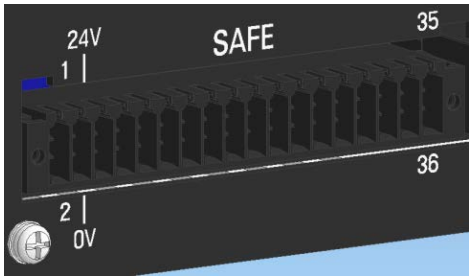


Cable side
Model: 10336-52A0-008
Manufacturer: 3M Company



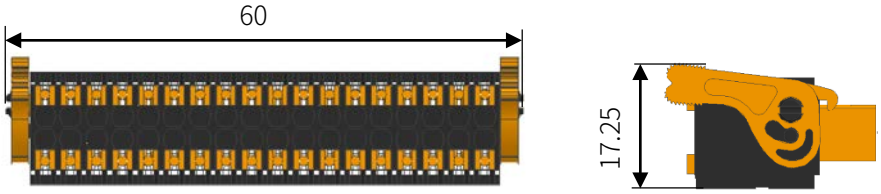
4.1.7.6 Safety input/output and I/O external power cable connector "SAFE" (Figure - ⑥)

Controller side
Model: S2C-SMT 3.50/36/90LF
Manufacturer: Weidmüller Interface GmbH & Co. KG



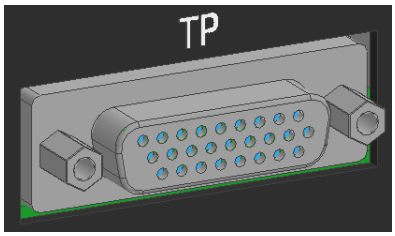
The mating connector is included with the controller. Refer to "4.1.4"SAFE" connector for connecting input/output external power supply and safety input/output (Figure 4.1-⑥)" or "4. Input/Output External Power Supply Cable and Safety Input/Output Cable" in the STEA1167_Interface Manual.

Cable side
Model: B2CF3.50/36/180LR
Manufacturer: Weidmüller Interface GmbH & Co. KG

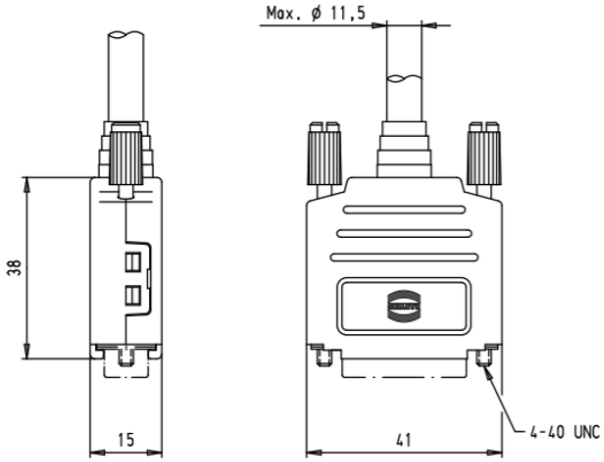


4.1.7.7 TP connector "TP" (Figure - ⑦)

Controller side
Model: 09 56 252 7613
Manufacturer: Harting Co., Ltd.



Cable side
Model: 09562005604
Manufacturer: Harting Co., Ltd.



4.2 Controller INPUT

4.2.1 Connector signal connection diagram

For a detailed drawing of the connector terminals, refer to "2. Connecting External Connection Cables" in the STEA1167_Interface Manual.

4.2.2 INPUT jumper connector

The following system input signal lines are provided as safety measures. Unless the following signal lines are jumpered, the robot will not operate. Therefore, the INPUT jumper connector in which the following system input lines are jumpered is included with this controller as standard. When this connector is used in TS5000, the internal jumper is set to P24G(-) Type P.

System input signal lines … INPUT-12 (STOP)
INPUT-14 (SVOFF)
INPUT-32 (BREAK)
INPUT-16 (P24V)
INPUT-17 (SYSINCOM)
INPUT-35 (P24G)

Jumpers of attached connector

INPUT (pin number)					
12-16	14-16	32-16	17-35	18-19[1)]	36-37[1)]

1) These jumpers are not used in this controller.
The robot can be operated without problem in this jumper setting.

Also, apply jumpers to the following signals if they are not used as system signals.

INPUT-13 (LOW_SPD)
INPUT-31 (CYCLE)
INPUT-16 (P24V)

MEMO

1. If the "STOP" signal is not jumpered, the robot automatic operation will not be available.

2. If the "SVOFF" signal and emergency stop contacts 1 and 2 are not jumpered, the servo power supply of the controller cannot be tuned on.

3. If the "BREAK" signal is not jumpered, the robot automatic operation will not be available.

4. If the "LOW_SPD" signal is not jumpered, the robot automatic operation speed will be slow.

5. If the "CYCLE" is not jumpered, the controller will switch to the cycle operation mode.

6. For the connectors supplied to the controller, 1. to 3. signals are already jumpered. However, note that the connection is P24G(—) common Type P. To use or change these signals, remove the connector jumper, and connect the wiring. Also, to operate the robot without using the system input signal, be sure to connect the supplied INPUT jumper connector to the INPUT connector on the controller.

7. If the customer creates the input signal line, change INPUT jumper connector wiring or arrange "13 INPUT Cable" in the STEA1167_TS5000 Interface Manual.

5. Operation Check

The cables must be connected by a qualified technician. Also, these processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

5.1 Controller operation check

・Startup check

Checking that the controller is starting normally.

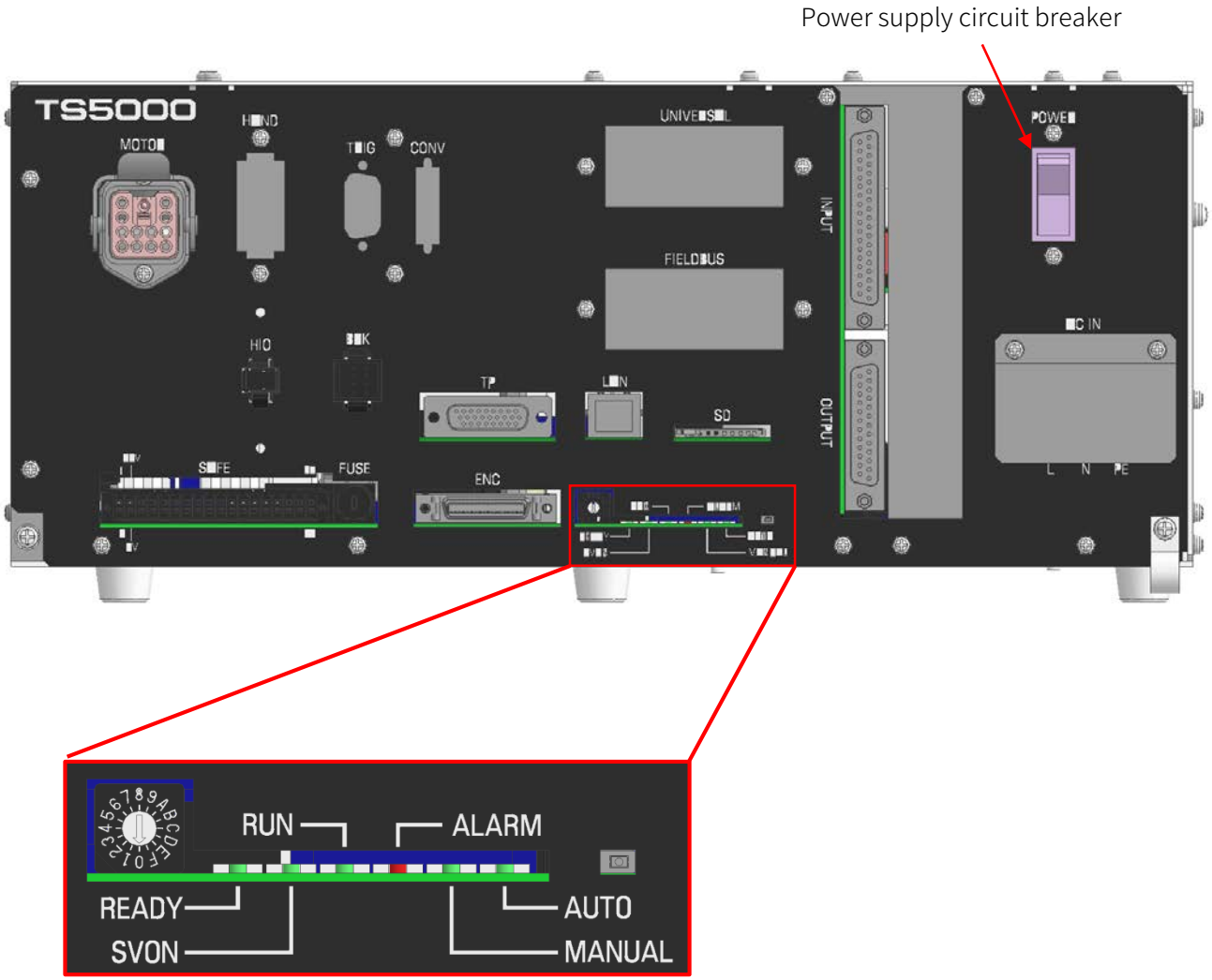
Before turning on the power of the controller, make sure that the ACIN is connected to the single-phase 200V and the 24V DC is connected to the SAFE connector, then push the power supply circuit breaker. When the controller starts normally, the “READY” LED indicator on the bottom front of the controller is turned on.

If the teach pendant is connected when the controller starts normally, the main screen will be displayed after startup.

・Reboot

Checking that the controller is rebooting normally.

To turn off the controller, push the power supply circuit breaker, then turn it on again. When the controller starts normally, the “READY” LED indicator on the bottom front of the controller is turned on.



LED indicator

5.2 Precautions for teaching operations

The teaching operations should be performed by qualified personnel. Be sure to follow the laws and regulations of each country.

⚠DANGER

- Be sure to perform the operation according to the instruction manual. Wrong handling could cause a collision with a robot or an electric shock.

- Be sure to perform teaching operation from outside the hazardous area to avoid being struck by a robot. Observe the following instructions when teaching in a hazardous area.

(1) The operation should be performed by at least two people. One person does the teaching while other one monitors from outside the hazardous area. Be sure to check on each other to prevent any misoperation.

(2) When teaching in a hazardous area, the operator must hold the master mode key and the teach pendant to prevent third-party operation.

(3) The operator should work in a way to always be able to push the emergency stop button. After confirming the operation area of the robot and surrounding shields, the operator works at a position where he can retreat immediately in the event of an abnormality.

(4) The supervisor monitors the operation from a safe position where the entire robot can be seen, and where the emergency stop button can immediately be pushed in the event of an abnormality. Also, the supervisor should keep other people away from entering the hazardous area. If an operator or another person does not follow the instructions of the supervisor, a collision with a robot could occur.

- If an error occurs when the device is turned on, or if the “READY” LED on the controller is not turned on, be sure to turn off the power to prevent electric shocks or fire.
- Be sure to perform daily inspections before operating the equipment. Failure to perform daily inspections could cause a collision with a faulty robot when it is operated. If the wires are short-circuited, a fire could occur.
- When turning on the servomotors, be sure to exit the hazardous area. There is a risk of being struck by a robot that moves unexpectedly.
- If a failure or malfunction occurs, be sure to stop the robot in an emergency and turn off the controller power. Otherwise, a fire could occur due to short-circuited wires or overheated core wires. Also, a collision with a robot in unexpected motion could occur.

⚠ WARNING

- Never enter a hazardous area during test operation as you may be hit by a robot.
- Never leave equipment in the work space that could interfere with your work and create a risk of stumbling or falling. When concentrating on the teaching operation, attention to the surroundings is neglected and the risk to crash into peripheral devices is increased.
- Never allow anyone other than the operator to approach the robot to prevent being hit by a robot.
- Never open the controller's cover during operation. Do not connect nor disconnect the connectors to avoid receiving an electric shock. Touching the connectors during operation increases the risk of being hit by a robot moving unexpectedly.
- Never change the data in the system configuration file during operation to avoid abnormal operations or crash of the robot.
- If an abnormal operation is confirmed during a teaching operation, be sure to stop using the robot as it may cause a serious accident.
- Be sure to stop operation if there is any abnormal noise, odor, or unexpected behavior from the controller or robot. If you ignore the abnormality and continue to operate, a fire could occur. Also, a collision with the robot in abnormal motion could occur.
- When teaching in the hazardous area, be sure to operate with the teach pendant or to hold the master key. If a robot is started by a third party during operation, there is a risk of being struck by a robot.
- Be sure to turn off the servomotors when leaving the workplace to avoid another operator who misunderstood that the robot stopped from being struck by a robot.
- If a failure or malfunction occurs, be sure to stop the robot in an emergency and turn off the controller power. Otherwise, a fire could occur due to short-circuited wires or overheated core wires. Also, a collision with a robot in unexpected motion could occur.
- When making the third axis servo-free, be sure to not put your hands or feet under the third axis. The third axis will fall and injure you by pinching your hands and feet.

⚠ CAUTION

- All hand output signals are turned off when the controller's power is turned off or when the power is turned on. Be sure to design the end effector so as not to drop objects due to a power failure. There is a risk of having the hands and feet pinched by the dropped end effector.

NOTICE

- Be sure to turn off the servomotors' power while the robot is stopped.

If the main power is turned off while the robot is operating, the arm may keep moving and collide with peripheral devices.

- When performing automatic operations, be sure to confirm the program execution status. Otherwise, there is a risk to interfere with peripheral equipment and cause the device to malfunction.

5.3 Precautions for operation

Operation should be performed by qualified personnel. Be sure to follow the laws and regulations of each country.

MEMO

1. *Do not change from MANUAL mode to AUTO mode while the safety protection functions are deactivated. When changing from MANUAL mode to AUTO mode, make sure that the safety protection functions are restored.*

⚠ DANGER

- Never operate the controller if there are damaged or missing parts as there is a risk of electric shocks or fire.
- Do not put conductive objects or fingers in the power supply terminal area to avoid receiving an electric shock.
- The operation mode of each controller should be displayed on the controller. If the operation mode is not known, a collision with a robot in unexpected motion could occur.
- Be sure to perform the operation according to the instruction manual. Wrong handling could cause a collision with a robot or an electric shock.
- If an error occurs when the device is turned on, or if the "READY" LED on the controller is not turned on, be sure to turn off the power as there is a risk of electric shocks or fire.
- When turning on the servomotors, be sure to exit the hazardous area to avoid being struck by a robot that moves unexpectedly.
- If a failure or malfunction occurs, be sure to stop the robot in an emergency and turn off the controller power. Otherwise, a fire could occur due to short-circuited wires or overheated core wires. Also, a collision with a robot in unexpected motion could occur.

⚠ WARNING

- Never open the controller cover during operation to avoid receiving an electric shock.
- Never change the data in the system configuration file during operation to avoid abnormal operations or crash of the robot.
- When using multiple controllers, be sure to select the same operation mode for all of the robots to avoid being struck by an unexpected robot.
- Do not enter a hazardous area during test operation to avoid being hit by a robot.
- Do not allow anyone other than the operator to approach the device to avoid being hit by a robot.
- Be sure to perform daily inspections before operating the equipment. Failure to perform daily inspections could cause a collision with a faulty robot when it is operated. If the wires are short-circuited, a fire could occur.
- Be sure to stop any operation if there is any abnormal noise, odor, or unexpected behavior from the controller or robot. If you ignore the abnormality and continue to operate, a fire could occur. Also, a collision with the robot in abnormal motion could occur.
- Do not enter the dangerous area of the robot during automatic operation to avoid being struck by a robot.
- Be sure to check the safety protection devices before starting any automatic operation. If the safety protection devices break down, it will not be able to perform an emergency stop, leading to the collision with the robot.

NOTICE

- Do not put obstacles within the robot movement range. Otherwise, a collision with a robot could occur.
- The output signal should be 0.1 A or less per signal. If a current exceeding the rated output is applied, the board, etc. may be damaged.
- The external I/O power supply should be of 24 V±10% (24 VDC). Applying a voltage exceeding the allowable value may damage the controller and cause a fire.
- Be sure to turn off the servomotors power while the robot is stopped.
  If the main power is turned off while the robot is operating, the arm may keep moving and collide with peripheral devices.
- When performing automatic operations, be sure to confirm the program execution status. Otherwise, there is a risk to interfere with peripheral equipment and cause the device to malfunction.

Maintenance

This part explains the maintenance and inspection of the controller.

6. Maintenance

The maintenance and inspection of the controller should be carried out only by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

Recommended protector:

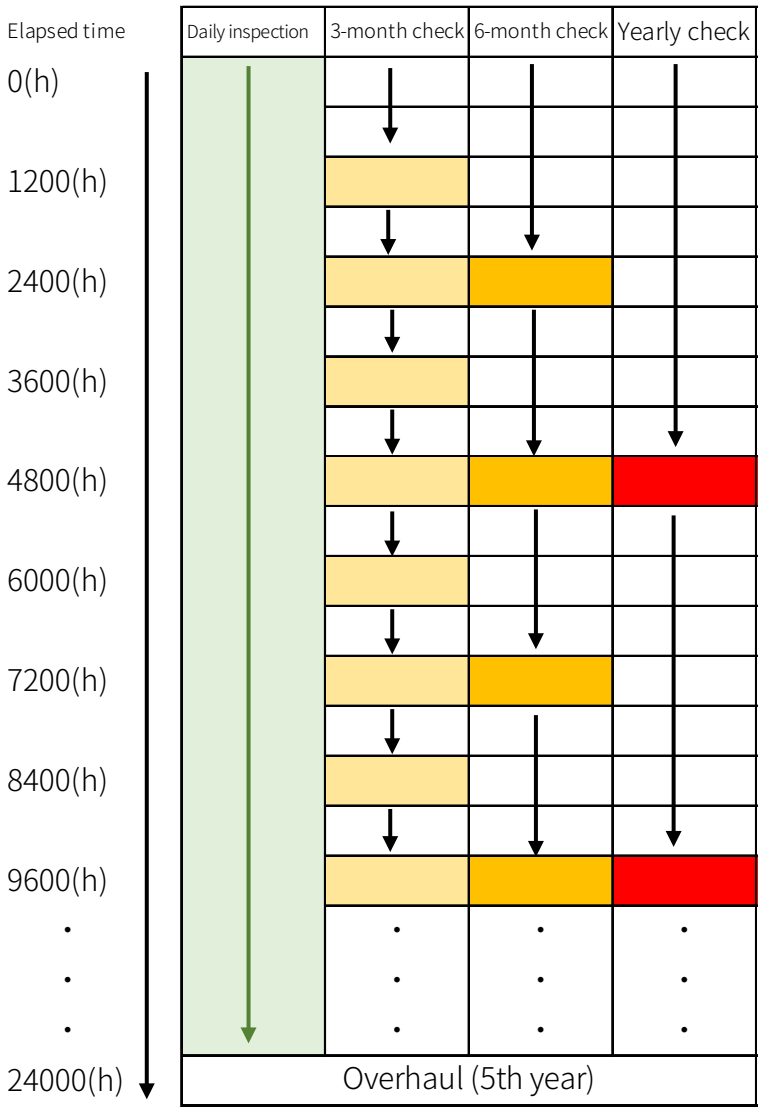
Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

⚠ WARNING

- Perform periodic maintenance and inspection. Failure to perform periodic inspection could result in a collision with a faulty robot when it is operated.

6.1 Maintenance Schedule

Daily and periodic maintenance and inspection are required. Inspection items are added in each periodic maintenance and inspection carried out every 1200 operating hours.



Periodic maintenance and inspection time calculation

16 operating hours per day: 1200 hours / 16 hours / 25 days = 3 months
24 operating hours per day: 1200 hours / 24 hours / 25 days = 2 months

If the operating hours per day are 16, maintenance and inspection are required every three months. If the operating hours per day are longer than 16, earlier maintenance and inspection are required.

Figure 6.1 Maintenance schedule

6.2 Maintenance/Inspection Items

Daily and periodic maintenance and inspection are required.
This section describes items of maintenance and inspection. For details of each maintenance and inspection, refer to the relevant section stated in the reference column.

6.2.1 Inspection in the power-off state

Table 6.1 Inspection in the power-off state

Inspection	Location	Daily check	3-month check	6-month check	Yearly check	Reference
Check screws for tightness. If they are loose, retighten them.	Cover mounting screws	○	○	○	○	7.6.4 7.6.5
Connector tightness	All connector section	○	○	○	○	4.1.1
Damage check Cleaning of accumulated dust	The entire robot controller	○	○	○	○	
Vent hole check	Vent holes	○	○	○	○	7.6.1

6.2.2 Inspection in the power-on state

Table 6.2 Inspection in the power-on state

Inspection	Location	Daily check	3-month check	6-month check	Yearly check	Reference
Emergency stop validation	Emergency Stop button	○	○	○	○	7.6.2 7.6.3
Enable switch validation	Enable switch	○	○	○	○	
Master mode switch MANUAL/AUTO mode validation	MANUAL/AUTO mode	○	○	○	○	1)
Robot/controller validation	All axis validation in MANUAL mode	○	○	○	○	

1) Refer to STEA1162_TP5000 Operation Manual.

6.2.3 Overhaul

Table 6.3 Overhaul

Maintenance details	Service parts	5 years	Reference
Battery replacement	Controller battery	○	7.6.4
Switching power supply replacement	Inside the controller	○	7.6.5

6.3 Maintenance Contract and Repair Service

6.3.1 Maintenance contract

Inspections after the six-month maintenance are difficult for customers to perform. Please request us to provide such inspections. We recommend that you make a service agreement with us at the time of purchase.

6.3.2 Repair service

If the system malfunctions, turn off the power supply, and contact our service department. Please notify us of the details of the failure and the following information stated on the nameplates of the robot and controller. (For the contact information of our service department, refer to "Service Network".)

- Controller model (MODEL)
- Controller's serial number (CONTROLLER S/N)
- Robot's serial number (ROBOT S/N)
- Controller's production date

For details on the serial number and manufacturing date, refer to the nameplate below on the robot base.

Also, refer to 1.5 Controller Model Name about controller model.

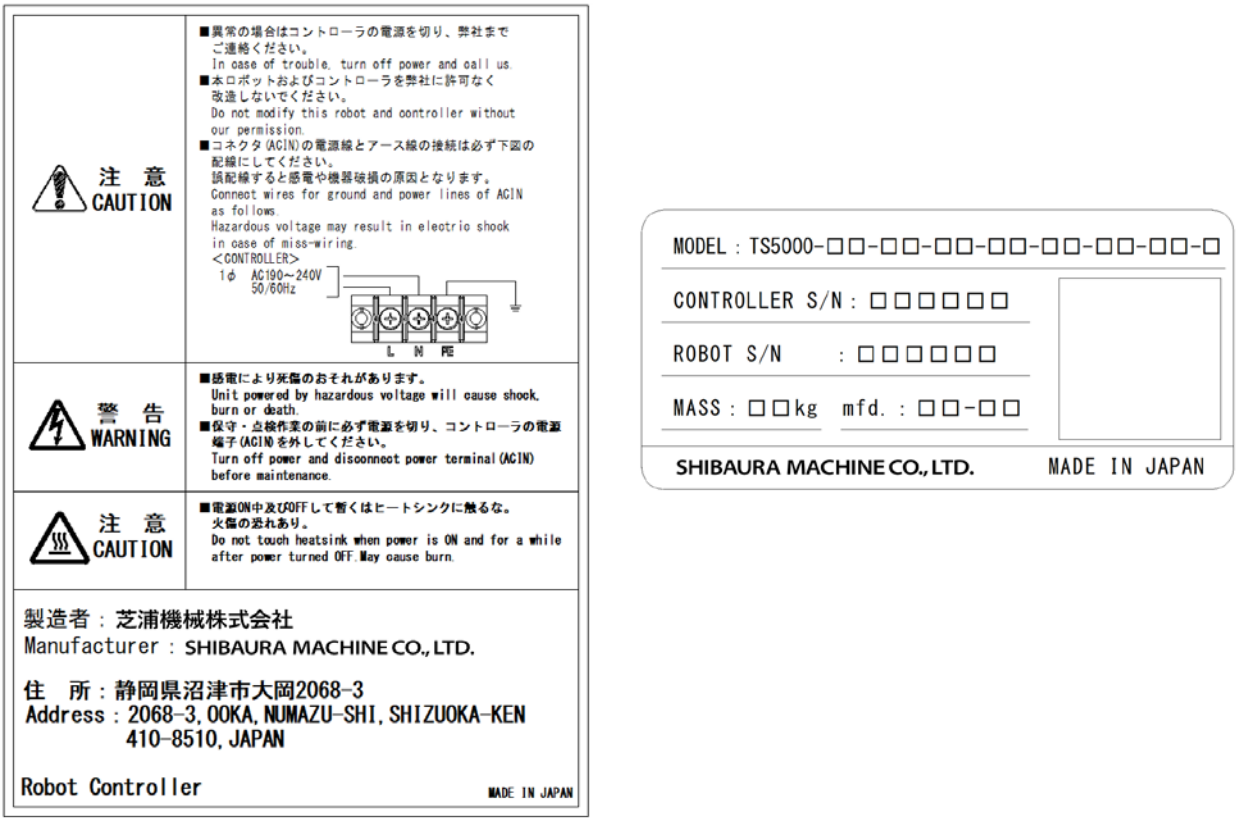


Figure 6.2 Nameplates

6.4 Modification

Be aware that SHIBAURA MACHINE shall not be liable for any modification performed by the customer.

⚠DANGER

- Never perform any part replacement or modifications other than those described in the instruction manual. Otherwise, reduced performance or a breakdown could lead to an electric shock, fire, or collision due to an unexpected robot motion.

6.5 Maintenance Tools

We recommend that you provide the following tools for maintenance.

- Phillips screwdrivers #1, #2, flat head screwdriver #2
- Nippers (to cut cable ties)

Recommended items other than tools

- Alcohol (for cleaning), cloth
- TSAssist (Program creation/teaching support software (option))

7. Maintenance of the Controller

The maintenance and inspection of the controller should be carried out only by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country. Also, no warranty is provided for any breakdowns or accidents that occur as a result of parts replaced by the customer.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

7.1 Precautions for Maintenance and Inspection

For maintenance and inspection of the controller, observe the following to ensure safety.

⚠DANGER

- When opening the cover of the controller, do not allow water or other fluids or foreign objects to get inside. This could cause a short-circuit inside the controller, which could lead to a fire or an electric shock due to ground leakage.
- Use SHIBAURA MACHINE special products for replacing certain maintenance parts. Wiring of unauthorized connectors and terminals could lead to a fire or an electric shock due to ground leakage.
- Be sure to request the SHIBAURA MACHINE servicing company if any maintenance or replacement work not described in this instruction manual is required. (A fee will be charged for this work.) SHIBAURA MACHINE will not be liable for any accidents or damage that occurs due to replacement work performed by the customer or its results.

⚠ WARNING

- Do not touch the controller terminals or internal board with your hands for about two minutes after the power cable is disconnected. A residual voltage is still in the large-capacity capacitor inside the controller, and this could result in an electric shock.
- Use SHIBAURA MACHINE authorized spare parts when replacing parts. Use of unauthorized parts could result in a fire or a collision with a robot in unexpected motion.
- Before starting any work, be sure to always wear protective gear, and unplug the power cable. If the power is turned on, the servo power module, servo amplifier module, switching power supply, and other components will carry a charge, and an electric shock could result.
- Turn off the power before opening the controller cover. If the cover is opened while the power is on, your hands or conductive tools could touch the power-carrying part, resulting in an electric shock.
- Perform the following actions whenever entering the movable range of the robot.
  - Wear protective gear
  - Place the teach pendant in an accessible location to the worker.
  - Change to MANUAL mode (remove the mode selector key so that the mode cannot be changed)

  ⇒ If the above operations are not performed, the emergency stop could be cleared by another person during the daily inspection work within the movable range, and a collision could occur with a robot that was set to continuous operation.
- Use voltage measurement and the LEDs to check that no PN voltage remains in the large-capacity capacitor. If a PN voltage remains, an electric shock could occur.
- Do not put conductive objects or fingers while the power is on. Touching the power-carrying part could result in an electric shock.

7.2 Controller Component Layout

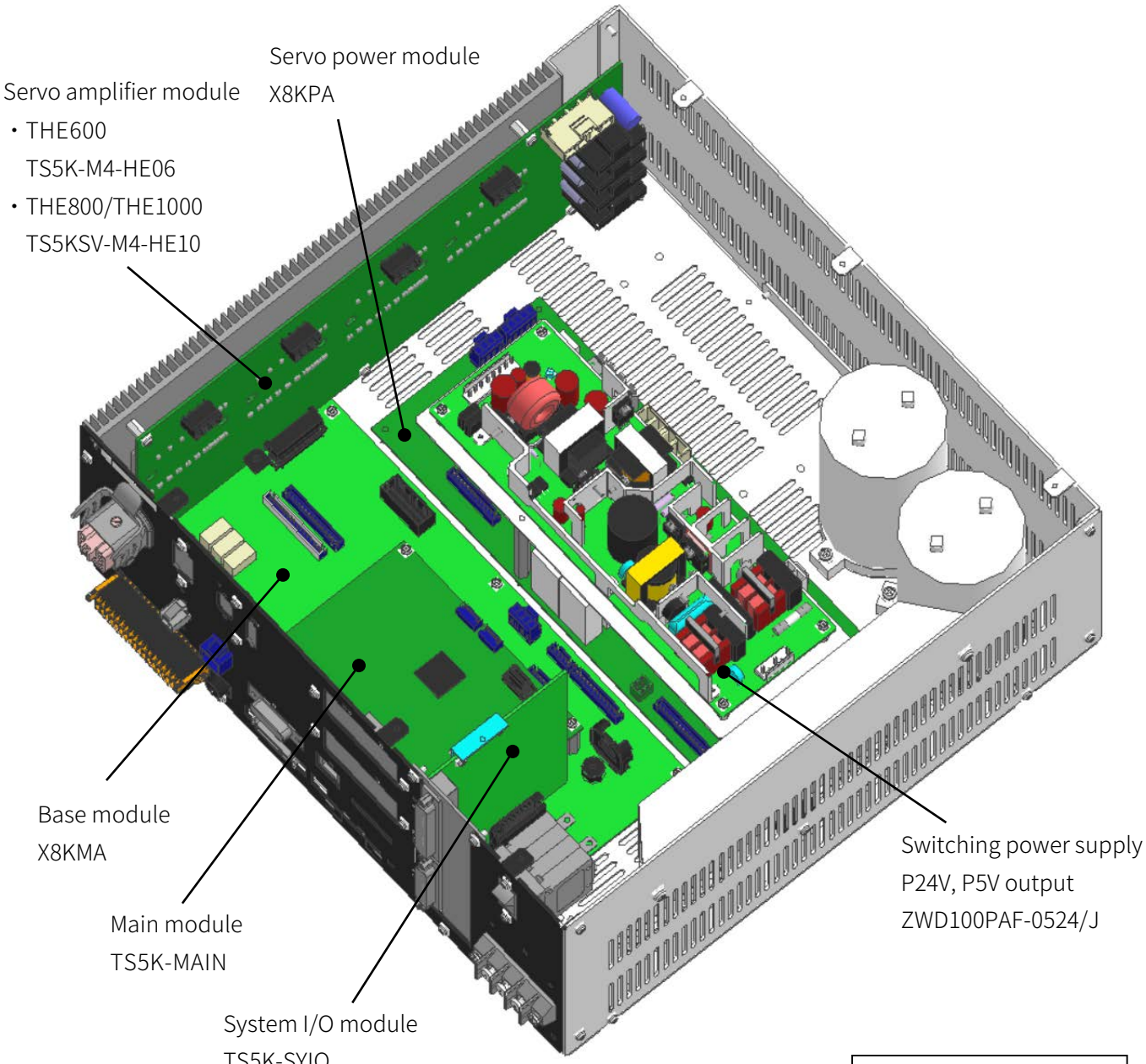


Figure 7.1 Controller component layout

7.3 TP5000's restricted operations

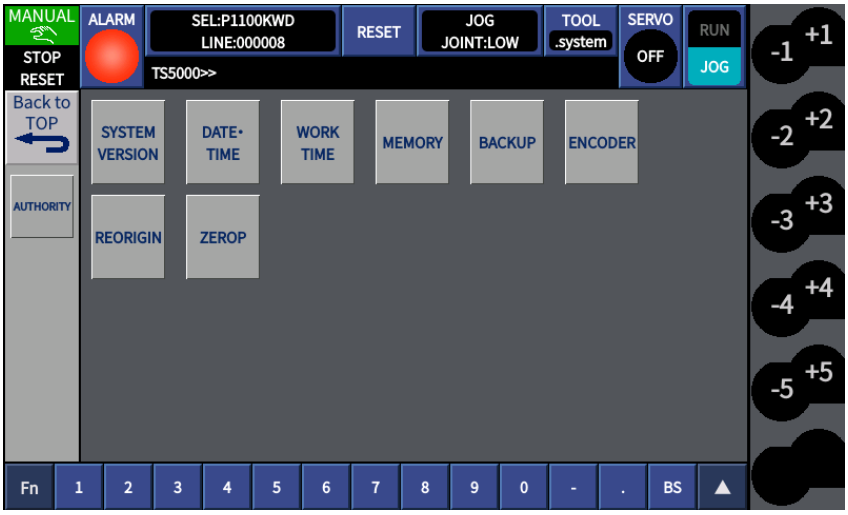
The teach pendant TP5000 has a screen operation restriction to prevent data loss due to incorrect operation by a third party.

To perform maintenance operations on the maintenance screen, the user type must be switched to "MAINTE".

＜Switching the user type to "MAINTE"＞

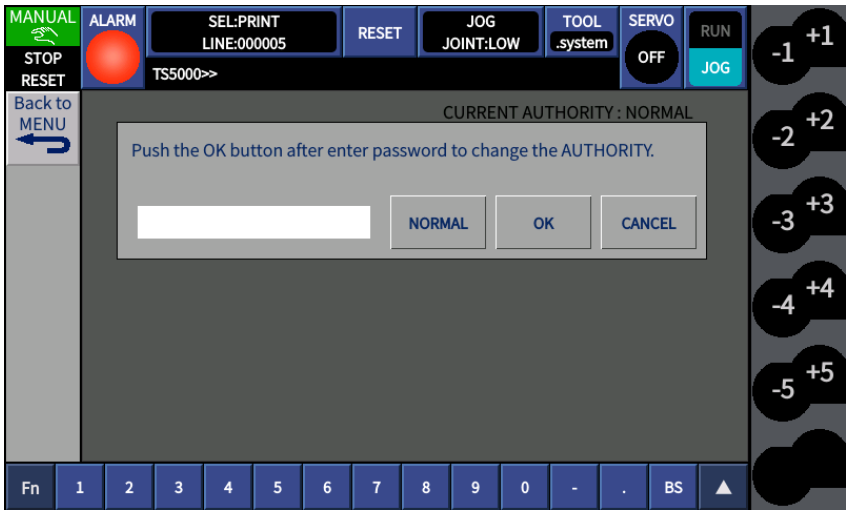
1 Press the "MAINTE" button on main screen

The maintenance menu screen is displayed.



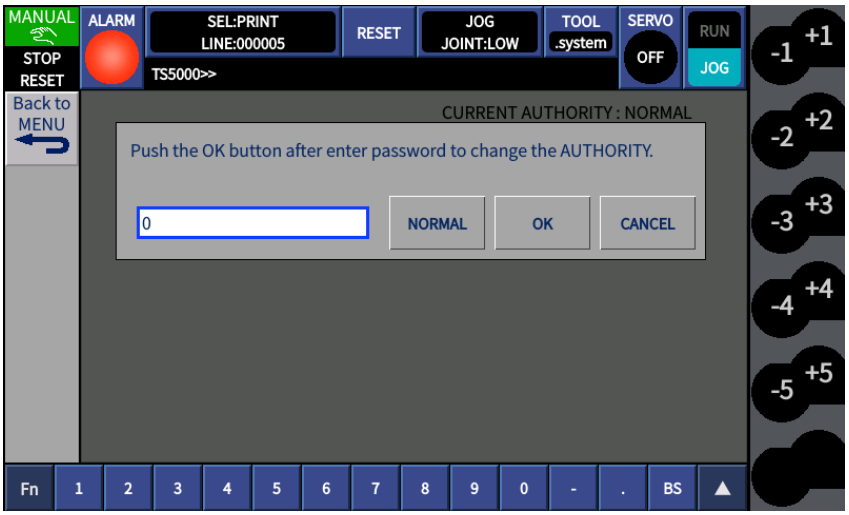
2 Press the "AUTHORITY"button.

The password entry screen is displayed.



3 Press the text box

A cursor appears in the text box to edit it.



4 Enter “0” using the keyboard and press the OK button.

The user type is changed to “MAINTE”.

MEMO

1. *The changed user type is valid until it is switched to another user type or until the TP5000 is turned off.*
2. *When the TP5000 is powered on, the user type is set to “NORMAL”.*

7.4 Data backup

Be sure to perform the controller maintenance after backing up the data.

7.4.1 Data Backup (TP5000)

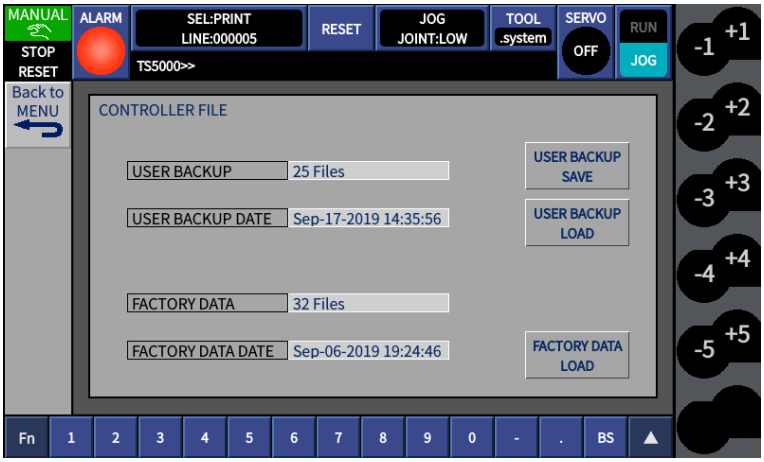
1 Insert the SD card where the TS5KBKUP folder was created into the TS5000 SD card slot after start.

2 Switch the user type to “MAINTE”.

For detailed operation, please refer to the 7.3 TP5000’s restricted operations section.

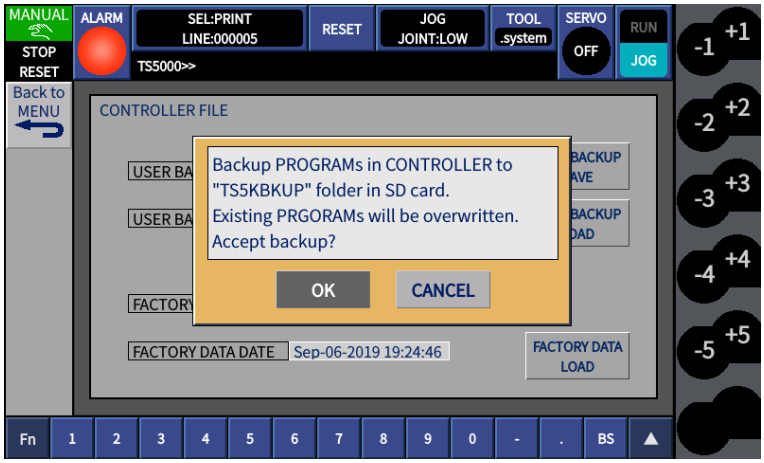
3 Press the “BACKUP” button on the maintenance screen.

The Backup screen is displayed.



4 Press the “BACKUP SAVE” button.

Confirmation of the backup operation menu is displayed.



5 Press the OK button.

The current controller program files and parameter files are saved in the TS5KBKUP folder on the SD card.

NOTICE

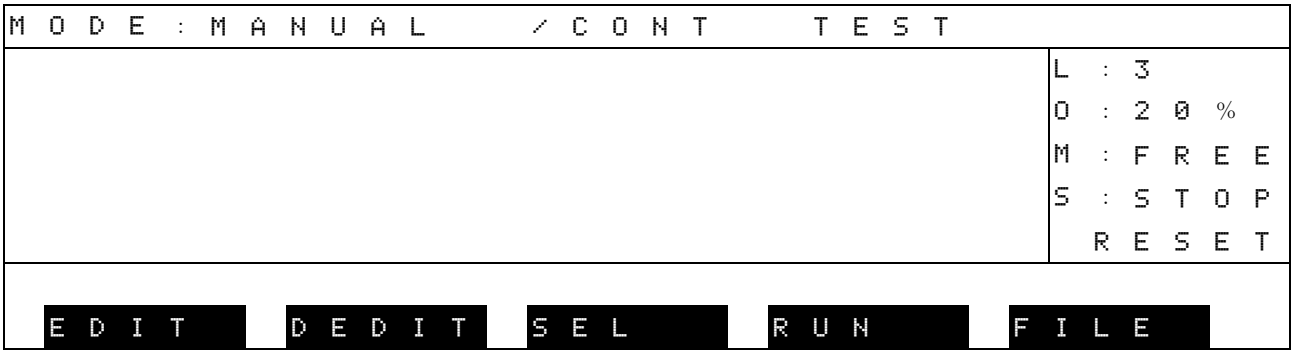
- Files in the TS5KBKUP folder on the SD card will be overwritten. To avoid overwriting, remove the target files from the TS5KBKUP folder and save it.

7.4.2 Data Backup (TP1000)

This section explains how to back up files (parameter files and program files) using the TP1000. For other operation methods using the TP1000, refer to the STEA1169_TP1000 operation manual.

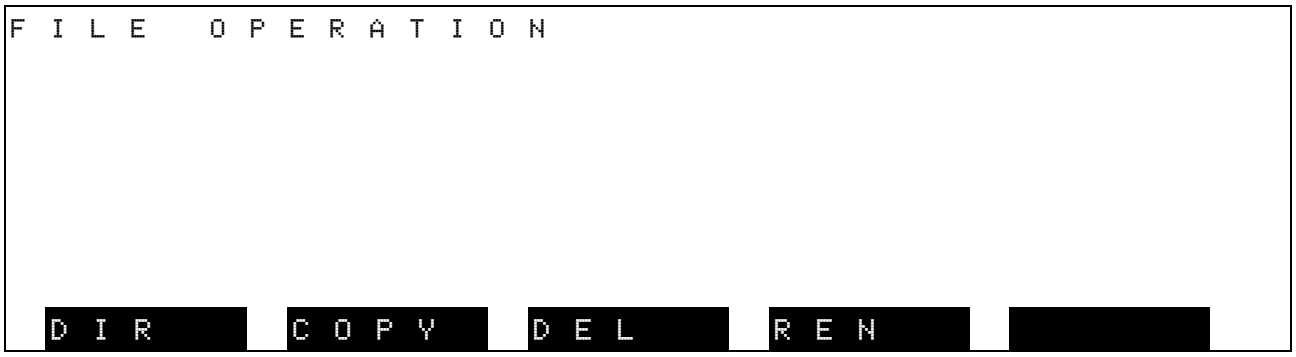
> *MEMO*
>
> *1. When using TP1000, data cannot be backed up to an SD card at once. Follow the procedure below for each required file.*

1 Press the NEXT key until "FILE" is displayed in the function menu on the main screen.



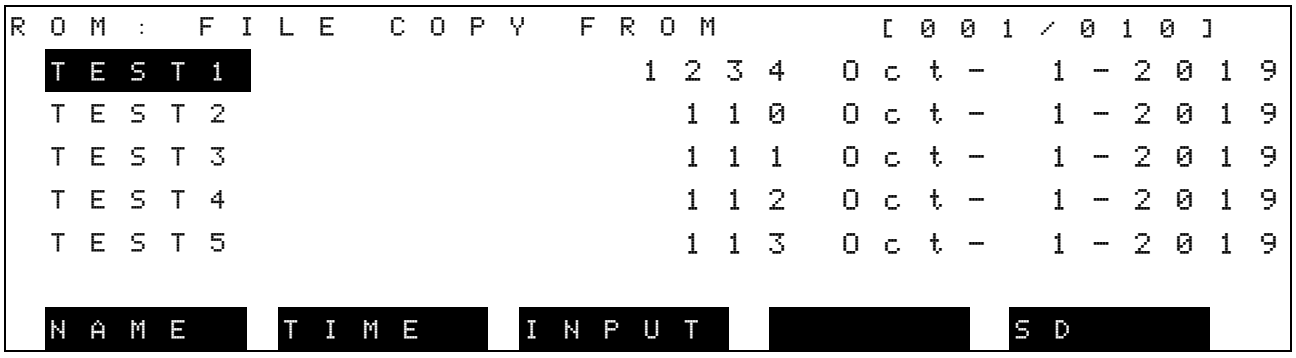
2 Press the FILE(F5) key.

The file operation screen is displayed.



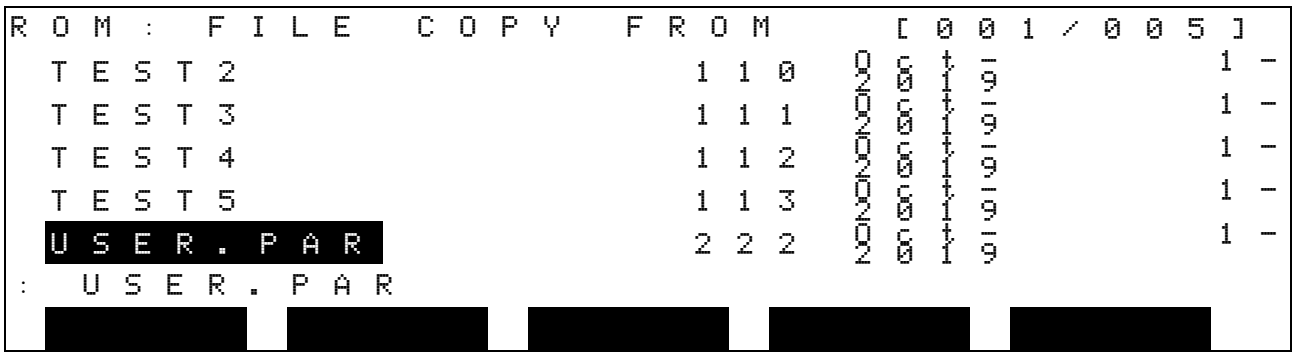
3 Press the COPY (F2) key on the file operation screen.

The operation file selection screen is displayed.



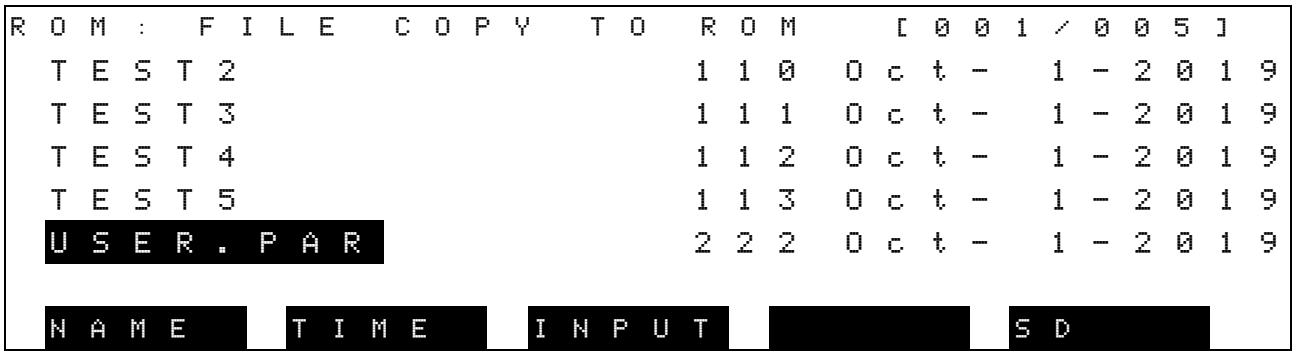
4 Move the cursor to the location of the file you want to copy and press the EXE key.

The name of the file is displayed on the command line.



5 Press the EXE key.

The operation file selection screen is displayed again.

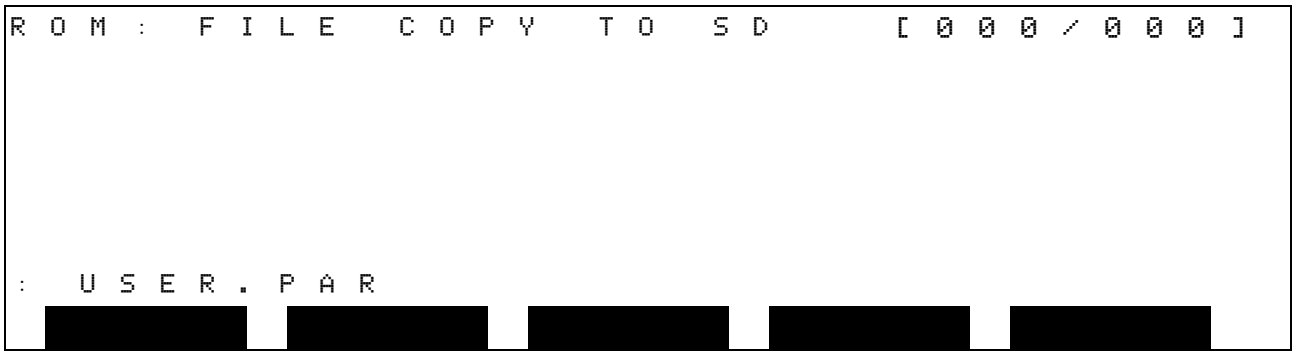


6 Press the SD/ROM(F5) key.

The operation file selection screen of the SD card (ROM) is displayed.

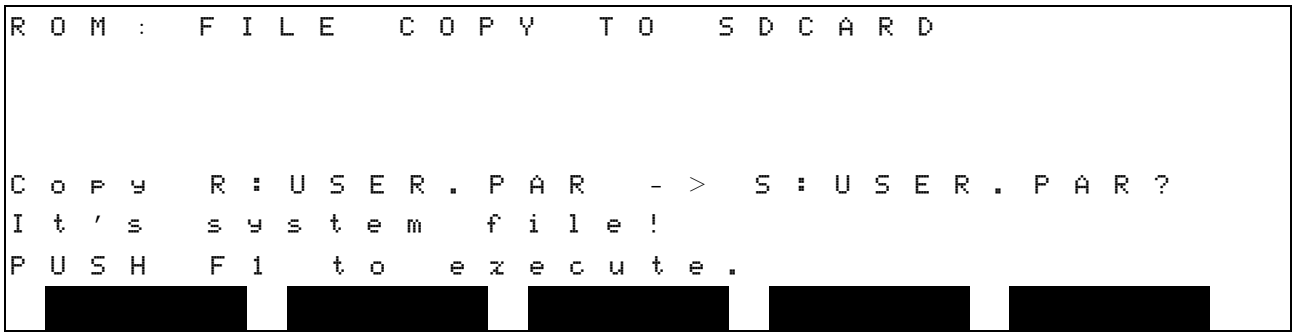
The name of the source file is displayed on the command line.

If the file to be copied is a system file, refer to step 7. Otherwise, refer to step “a”.



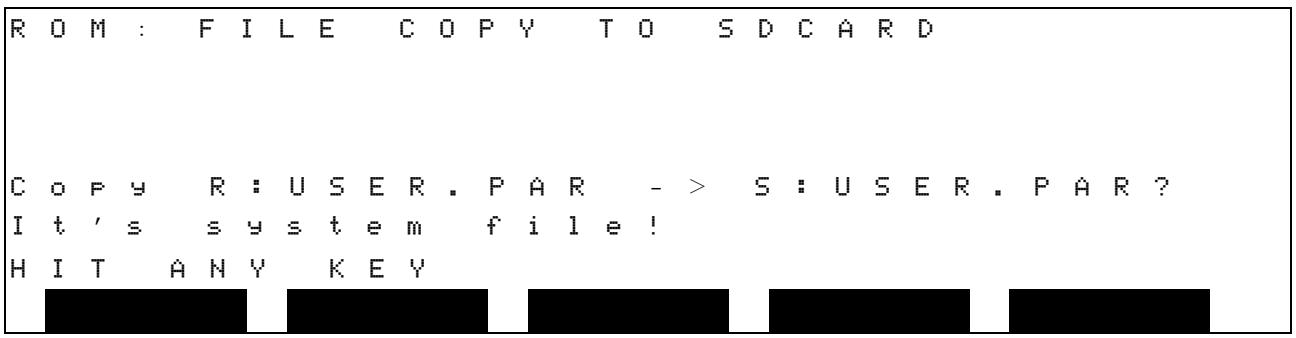
7 Confirm the copy destination file name and press the EXE key.

A confirmation message is displayed.



8 Press the F1 key.

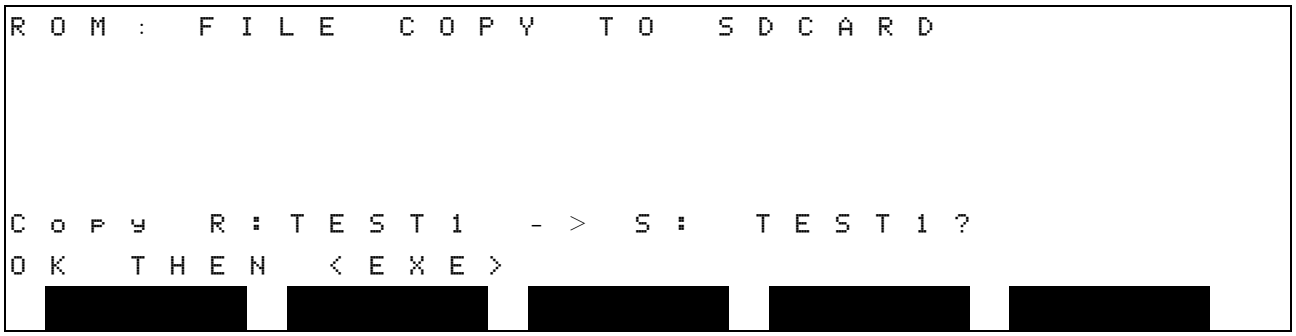
If there is no existing file with the copy destination file name, file copy will be done.



The following screen is displayed when TEST1 (a program file) is selected in the step 6.

a Confirm the copy destination file name and press the EXE key.

A confirmation message is displayed.



b Press the EXE key.

If there is no existing file with the copy destination file name, file copy will be done.



7.5 Restoring data

This section explains how to restore your data.

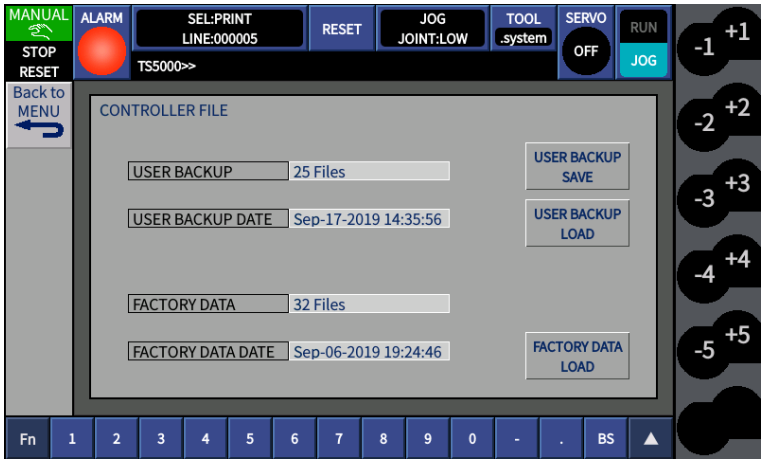
7.5.1 Data Restoration (TP5000)

This section explains how to restore data saved in Section 7.4.1.

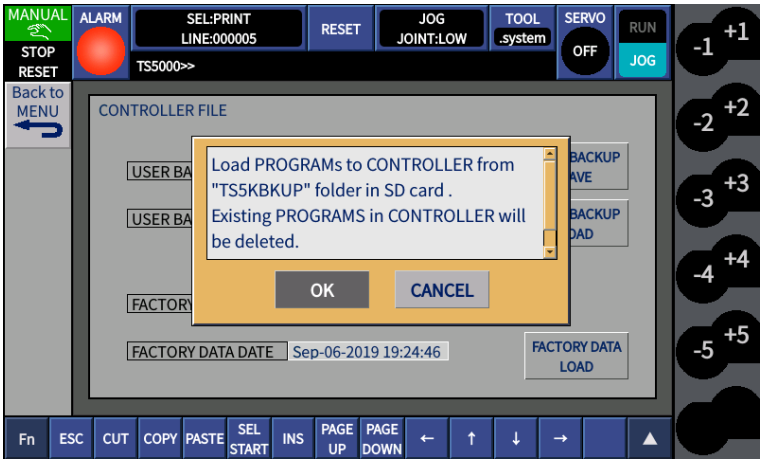
1 Insert the SD card that was used for backup into the TS5000 SD card slot after start.

2 Switch the user type to “MAINTE”.
For detailed operation, please refer to the 7.3 TP5000’s restricted operations section.

3 Press the “BACKUP” button on the maintenance screen.
The Backup screen is displayed



4 Press the “BACKUP LOAD” button.
Confirmation of the loading operation menu is displayed.



5 Press the OK button.
All the files in the controller will be deleted and the files located in the TS5KBKUP folder in the SD card will be loaded.

NOTICE

●USER BACKUP LOAD deletes all files in the controller. Be sure to first save the files inside the controller on an external device using file transfer.

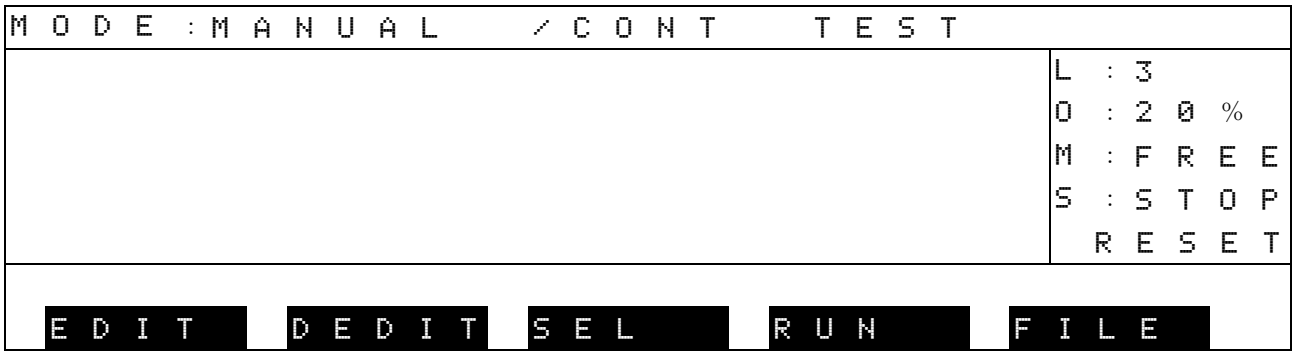
7.5.2 Data Restoration (TP1000)

This section explains how to restore the data backed up in 7.4.2.

MEMO

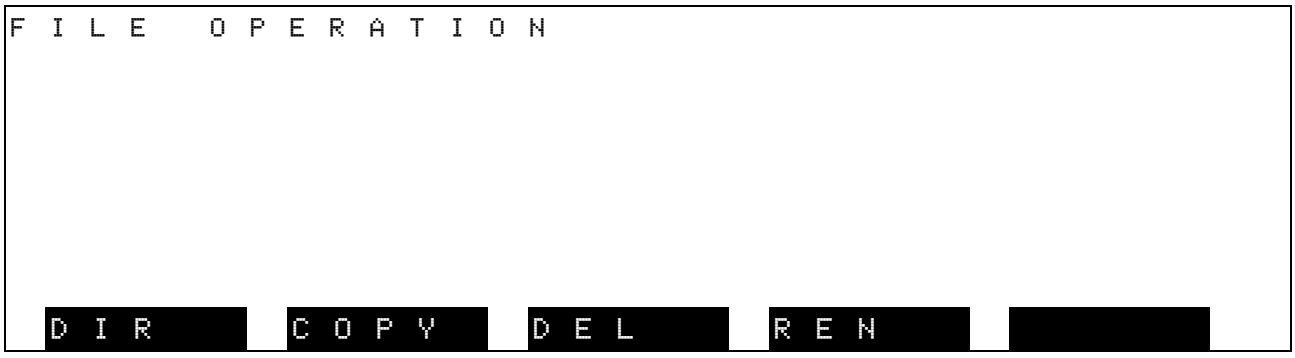
1. *When using TP1000, data cannot be restored from an SD card at once. Follow the procedure below for each required file.*

1 Press the NEXT key until "FILE" is displayed in the function menu on the main screen.



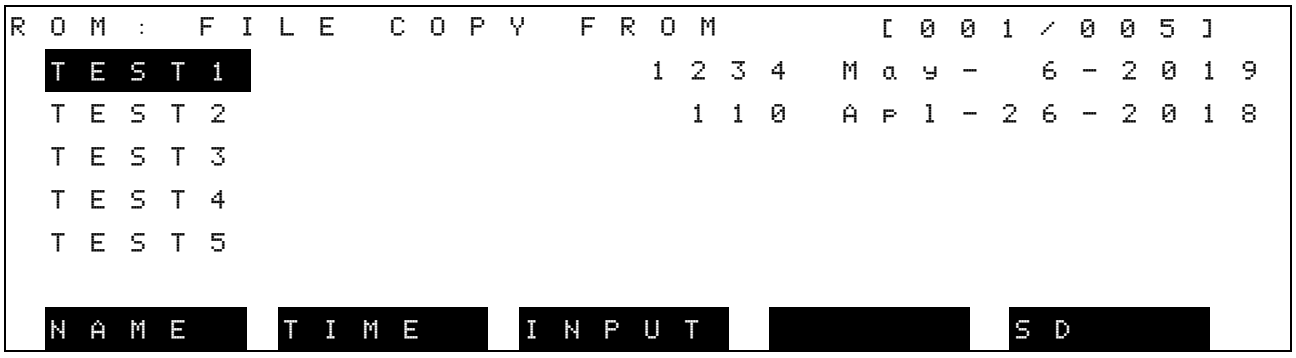
2 Press the FILE(F5) key.

The file operation screen is displayed.



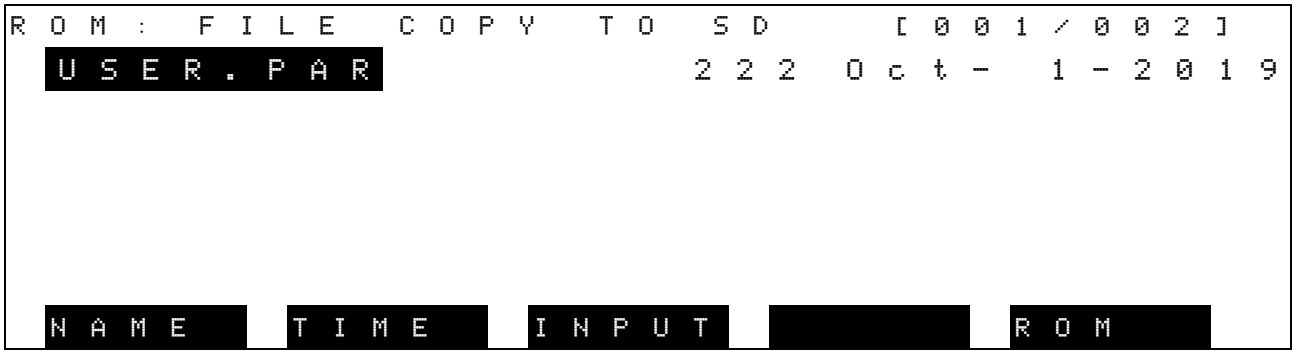
3 Press the COPY (F2) key on the file operation screen

The operation file selection screen is displayed.



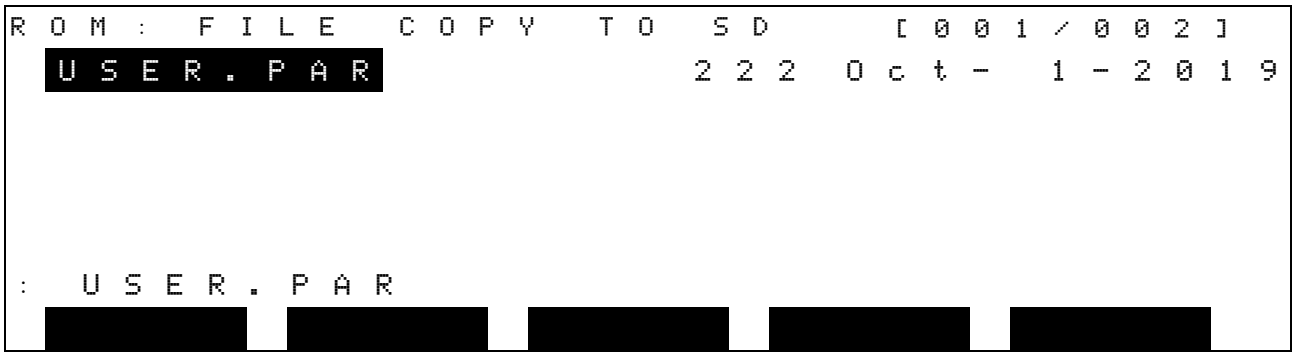
4 Press the SD/ROM(F5) key.

The operation file selection screen of the SD card (ROM) is displayed.



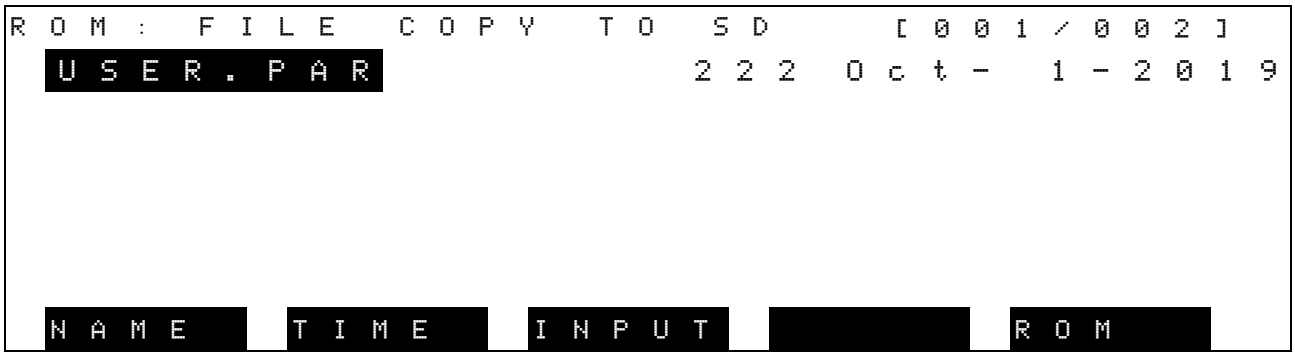
5 Move the cursor to the location of the file you want to copy and press the EXE key.

The name of the file is displayed on the command line.



6 Press the EXE key.

The operation file selection screen is displayed again.



7 Press the SD / ROM (F5) key.

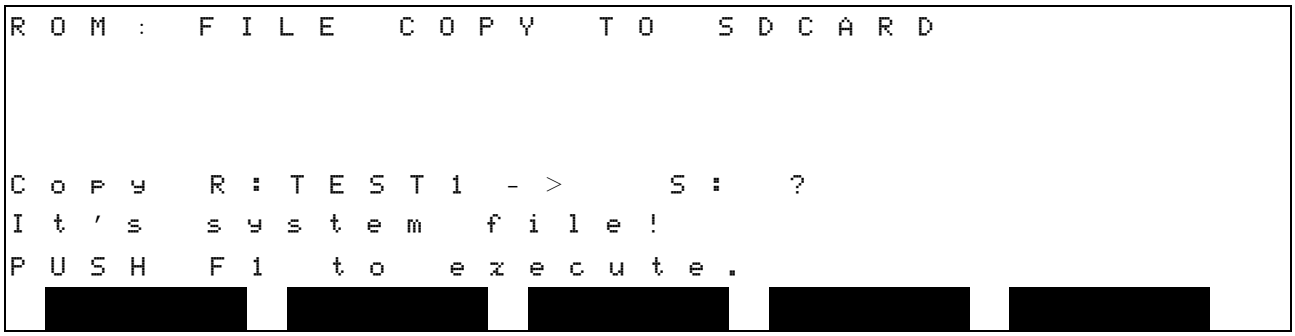
The operation file selection screen in the SD card (ROM) is displayed.

The name of the source file is displayed on the command line.



8 Confirm the copy destination file name and press the EXE key.

A confirmation message is displayed.



9 Press the F1 key.

If there is no existing file with the copy destination file name, file copy will be done.



7.5.3 Restoration to the factory settings

This section shows how to restore the setting to factory.

> *MEMO*
>
> 1. *When this procedure is executed, the controller status returns to the factory settings. Be sure to back up the necessary parameters and programs that have been changed by referring to 7.4 Data backup.*

1 Create a TS5KBOOT folder on SD card and store the FACTBKUP.BIN file.

TS5KBOOT folder
└──── FACTBKUP.BIN

> *MEMO*
>
> 1. *Use an SD card formatted with FAT16 or FAT32.*
> 2. *FACTBKUP.BIN is stored on the attached DVD.*

2 Insert the SD card into the power-off controller.

3 Turn on the power and start the controller.

The following screen is displayed only when the TP1000 is connected.

```
TS5000 RSW:0 System Update
X8KCB-01B 2019-10-01 00:00 AAAA
S068B     2019-10-01 00:00 AAAA
X8KSB-01B 2019-10-01 00:00 AAAA
X8KCRB-01B 2019-10-01 00:00 AAAA
X8KCRX-01B 2019-10-01 00:00 AAAA
MAC ADDR : 02.00.00.00.00.01
   Press the push switch to start
```

4 After startup, if there is repetitive sound alarm, press and hold the maintenance switch for about 3 seconds.

The alarm will go off because the file is being written.

5 After writing the BIN file, if there is repetitive sound alarm turn off the power.

Restoration to factory settings is complete.

7.6 Maintenance Procedure

7.6.1 Checking the vent holes of the controller

If the vent holes are blocked, the temperature inside the controller increases, resulting in malfunction. Check the following to ensure that the vent holes are not blocked.

- Remove any object in front of the vent holes to ensure ventilation.
- Ensure there is no substance stuck in the vent holes.
  Remove any substance stuck in the vent holes.

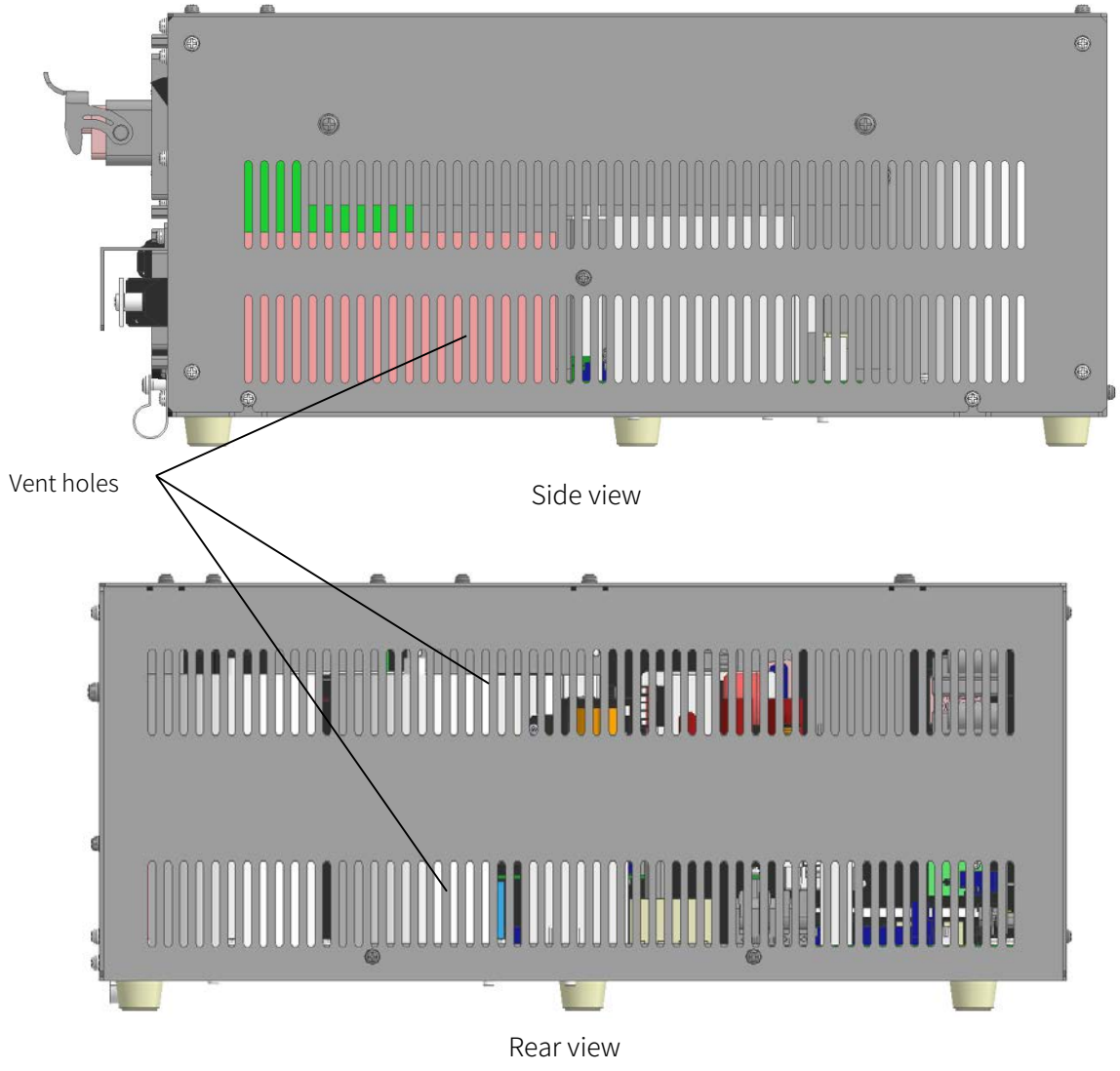


Figure 7.2 Side view of the controller

7.6.2 Safety device validation (TP5000)

Confirm that the emergency stop switch and other switches of the teach pendant and safety devices connected to the external operation input signals function normally.

Confirm that the emergency stop switch on the teach pendant works normally.

1. **Turn the master mode switch① on the front of the TP5000 to the MANUAL position.**
2. **Turn the emergency stop switch ② in the clockwise direction to confirm that the emergency stop is turned off.**
3. **While holding the Enable switch ③ of the teach pendant at its neutral position, press the "Servo ON" button on the teach pendant screen to turn on the servo. Confirm that the "SVON" LED on the front of the controller was turned on and that the screen changed.**
4. **Press the emergency stop switch ② on the teach pendant and confirm that the servo is turned off. Confirm that the "SVON" LED on the controller front was turned off and that the screen changed.**
   * In this case, confirm that the emergency stop switch ② is in the pressed state.
5. **Turn the emergency stop switch ② clockwise and confirm that it turns off.**

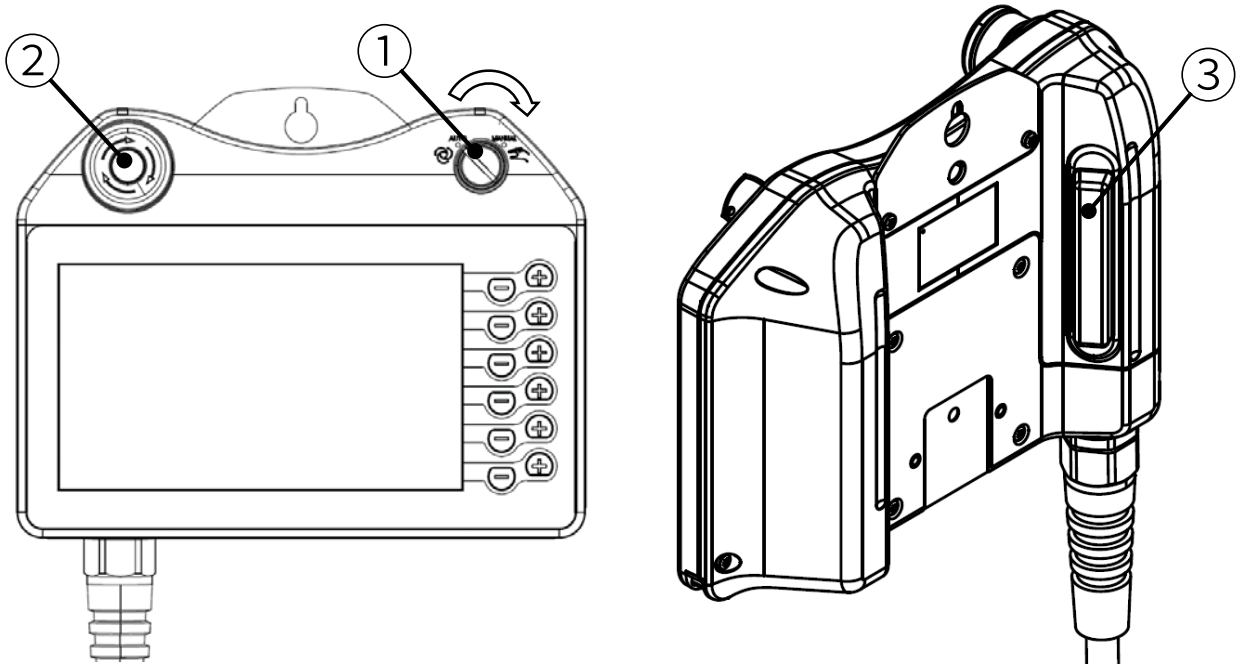


Figure 7.3 Emergency stop switch of the teach pendant (TP5000)

Confirm that safety devices connected function normally.

1. **Connect the power plug to the power supply, turn on the servo, turn on the safety devices (external emergency stop switch, beam type safety device, foot switch, etc.) connected to the "emergency stop" external operation signals, and confirm that the servo can be turned off.**
2. **Confirm that safety devices connected to the "low speed instruction" external operation input signals turn on and function normally.**

7.6.3 Safety device validation (TP1000)

Confirm that the emergency stop switch and other switches of the teach pendant and safety devices connected to the external operation input signals function normally.

Confirm that the emergency stop switch of the teach pendant functions normally.

1. **While in MANUAL mode, turn the emergency stop switch ① in the clockwise direction to confirm that the emergency stop switch ① is turned off.**
2. **While holding the Enable switch ③ of the teach pendant at its neutral position, press the "Servo ON" switch ② to turn on the servo. Confirm that the LED of the Servo ON switch ② is turned on.**
3. **Press the emergency stop switch ① of the teach pendant and confirm that the servo is turned off. The LED of the Servo ON switch ② was turns off.**
   * In this case, confirm that the emergency stop switch ① is in the pressed state.
4. **Turn the emergency stop switch ① clockwise and confirm that it turns off.**

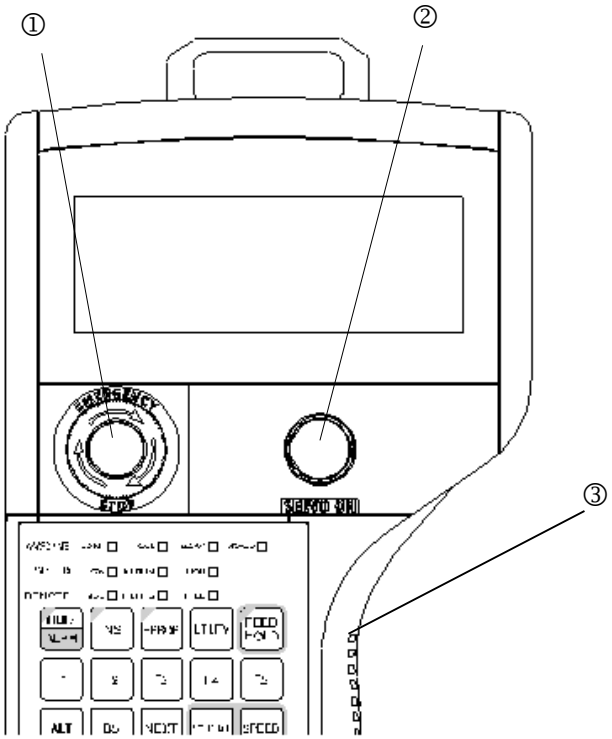


Figure 7.4 Emergency stop switch of the teach pendant (TP1000)

Confirm that safety devices connected to the external operation signals function normally.

1. **Connect the power plug to the power supply, turn on the servo, turn on the safety devices (external emergency stop switch, beam type safety device, foot switch, etc.) connected to the "emergency stop" external operation signals, and confirm that the servo can be turn off.**
2. **Confirm that safety devices connected to the "low speed instruction" external operation input signals function normally.**

7.6.4 Replacing the controller battery

Controller maintenance and inspection must be performed by qualified personnel. These processes must be compliant with the laws, regulations, and ordinances of the respective country. Replacement work can be performed by the SHIBAURA MACHINE service department for a fee. Also, no warranty is provided for any breakdowns or accidents that occur as a result of parts replaced by the customer.

⚠DANGER

- Do not attempt to incinerate, disassemble, short-circuit, or charge the battery. This could cause the battery to rupture, resulting in a fire. Dispose of the battery in compliance with the regulations of the customer.

The TS5000 uses button cell batteries for running the realtime clock (RTC).
The controller battery should be replaced about every 5 years. After the button cell battery is used for a certain period of time, the battery voltage will drop, the date and time information will be lost, and in certain cases, battery fluid could leak, resulting in a breakdown. After replacing the battery, set the date and time.
Also, do not remove the button cell battery except when replacing it. Otherwise, the date and time data saved internally will be lost.

Replace the battery if any of the following alarms occurs.
"MAIN Board Battery alarm" (Alarm code 1-145)

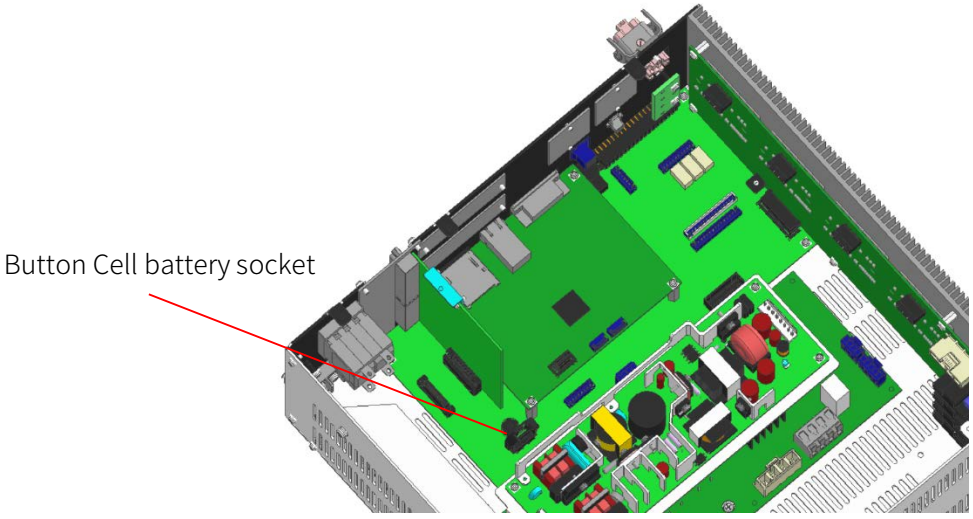


Figure 7.5 Lithium battery location

Table 7.1 Battery replacement

Button cell battery type	Manufacturer	Battery life
CR2032H	Maxell Holdings, Ltd.	Follow the manufacturer's recommendations

Replacement procedure

1 **Turn off the circuit breaker for the power supply, and unplug the power cable.**

2 **Remove the power cable from the ACIN connector.**

3 **Obtain a new button cell battery, remove the SEMS screws (M3×8×7) securing the cover, and remove the cover from the controller body.**

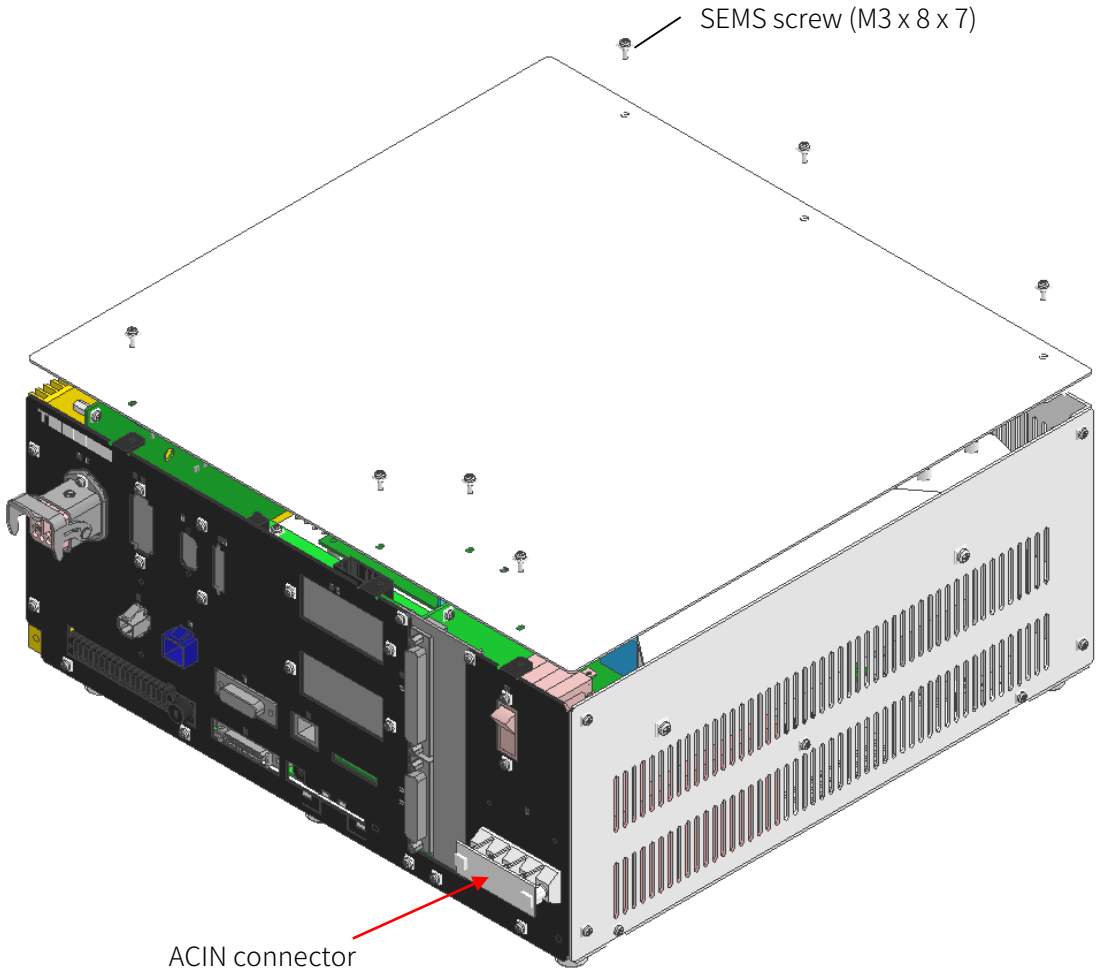


Figure 7.6 Removing the cover

4 **The controller battery is located at the position shown in Figure 7.5.**
To remove the button battery, push the button battery in the direction of ① as shown in Figure 7.7, and then remove it upward in the direction of ② so that it does not get caught in the claws on the top. When inserting the button battery, reverse the procedure.
*** Even if the button cell battery was not used, its capacity is reduced due to self-discharge. Use a button cell battery that was purchased within the last three years (stored at room temperature).**

5 **After installing the button cell battery, use the SEMS screws (M3×8×7) to remount the cover to the controller body. (*After replacing the battery, remount the cover so that there is no looseness or rattling of the screws.)**

6 **Set the date and time when the controller is started for the first time after battery replacement.**

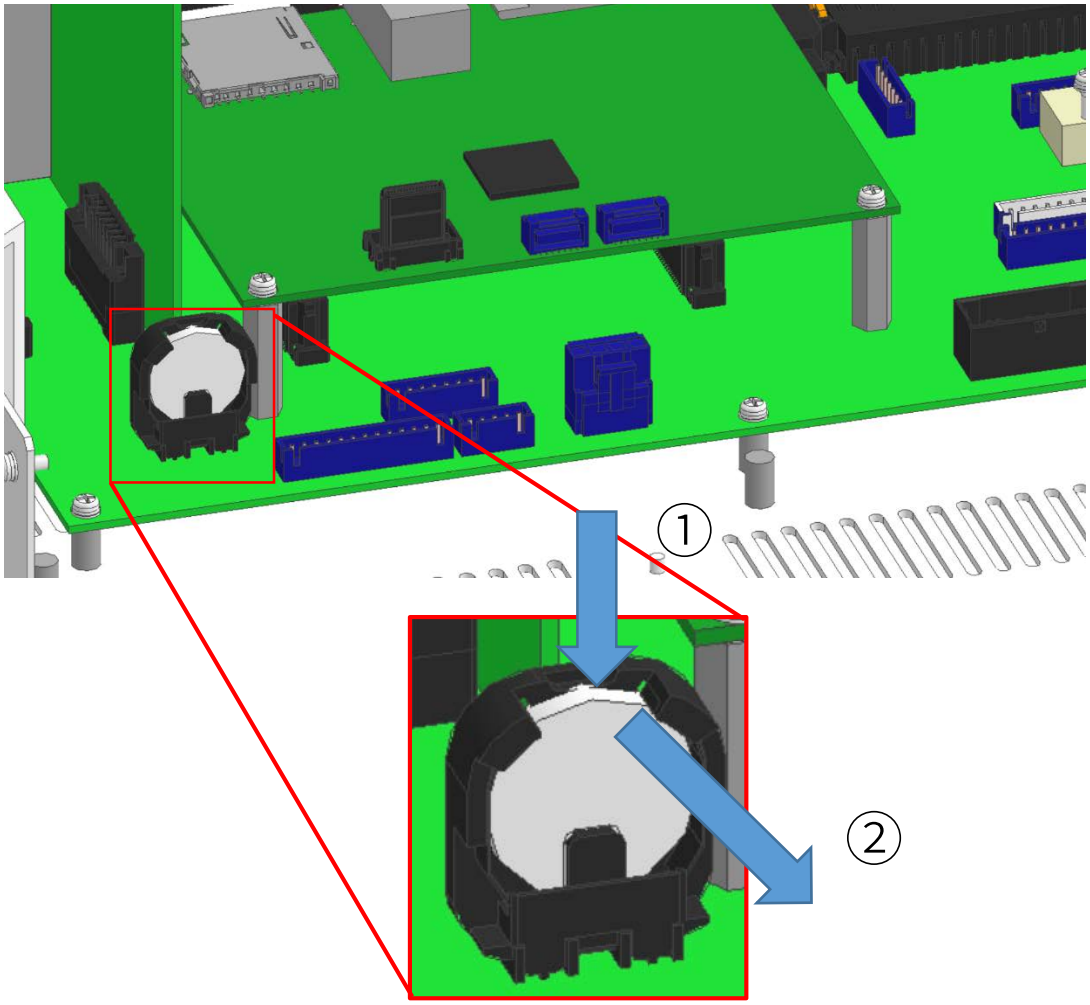


Figure 7.7 Removing the button cell battery

7.6.5 Switching power supply replacement

Controller maintenance and inspection must be performed by qualified personnel. These processes must be compliant with the laws, regulations, and ordinances of the respective country. Replacement work can be performed by the SHIBAURA MACHINE service department for a fee. Also, no warranty is provided for any breakdowns or accidents that occur as a result of parts replaced by the customer.

⚠DANGER

- Install the switching power supply in the location shown in the instruction manual. If the up and bottom or right and left sides are backwards, or the connector is different, this could cause a short-circuit or broken wire, resulting in a fire or an electric shock due to ground leakage.

The lifespan of the switching power supplies used for the controller depends on the use condition.
The lifespan of the power supply may extend from 5 years to 10 years depending on the use condition, but we recommend replacing the power supply every five years as preventive maintenance.

Replacement Procedure

1 **Turn off the breaker for the power supply, and remove the power plug from the power supply.**

2 **Remove the power cable from the ACIN connector.**

3 **Use a Phillips screwdriver #2 to remove the SEMS screws (M3×8×7) securing the cover, and remove the cover from the controller body.**

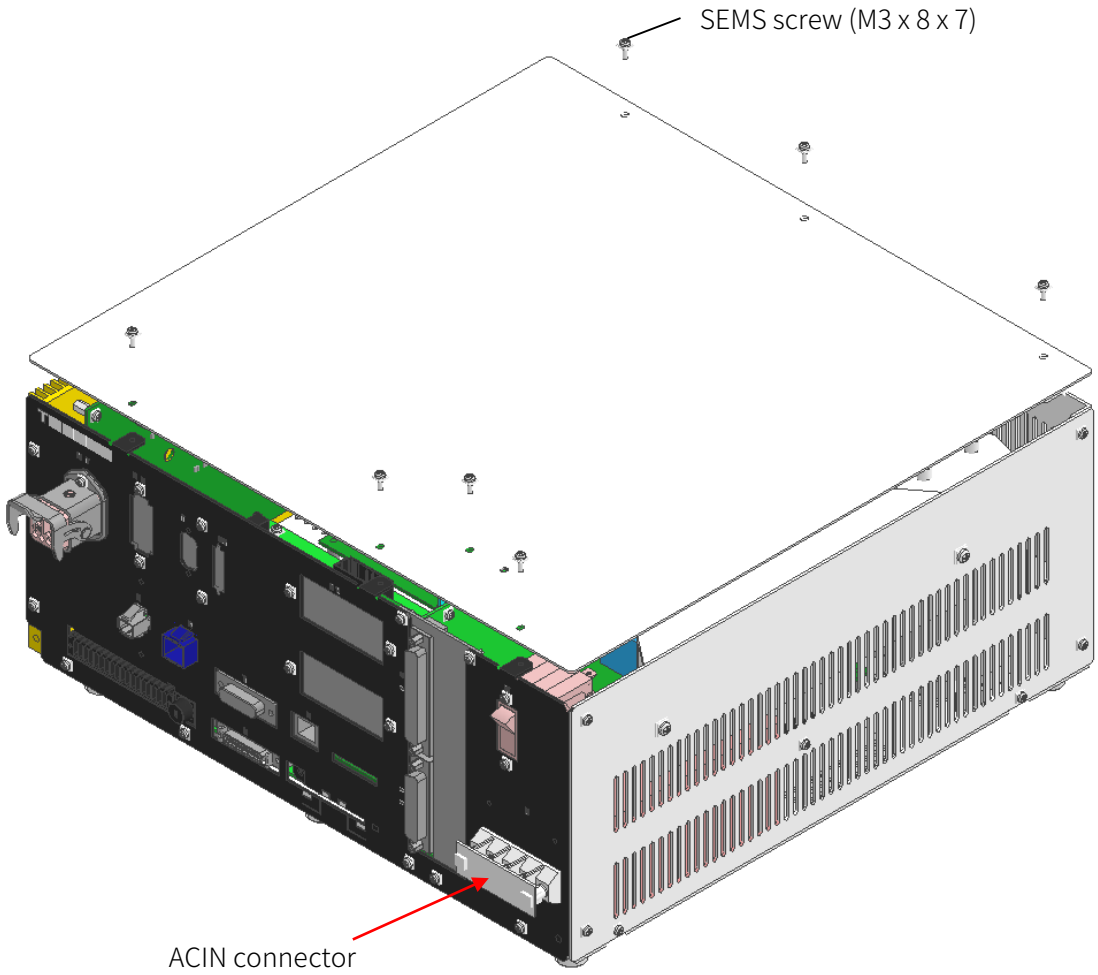


Figure 7.8 Removing the cover

4 **Remove the TS5KP-DC-KM cable (switching power supply secondary side) and the SW cable (switching power supply primary side) connected to the switching power supply as showed in Figure 7.10.**

5 **Use a Phillips screwdriver #2 to remove the SEMS screws (M3×8×4), and replace the switching power supply.**

6 **After replacing the switching power supply, reconnect the TS5KP-DC-KM cable (switching power supply secondary side) and the SW cable (switching power supply primary side).**

7 **Use the SEMS screws (M3×8×7) to remount the cover to the controller's body. (*After replacing the switching power supply, remount the cover so that there is no looseness or rattling of the screws.)**

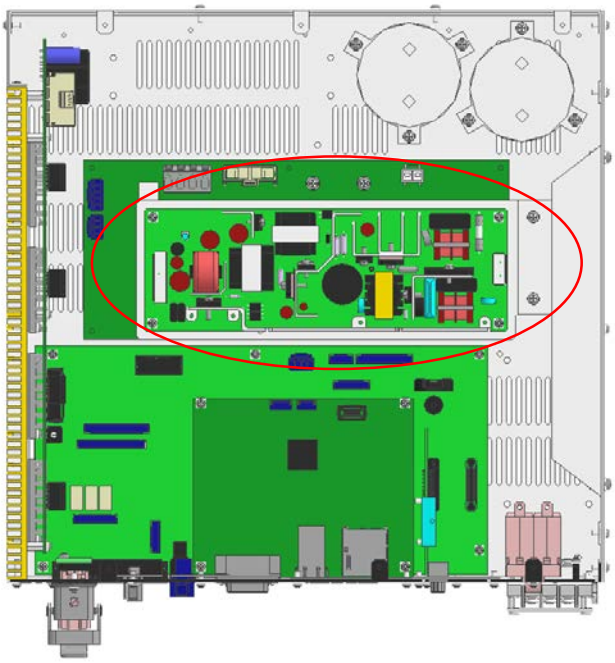


Figure 7.9 Controller internal front side

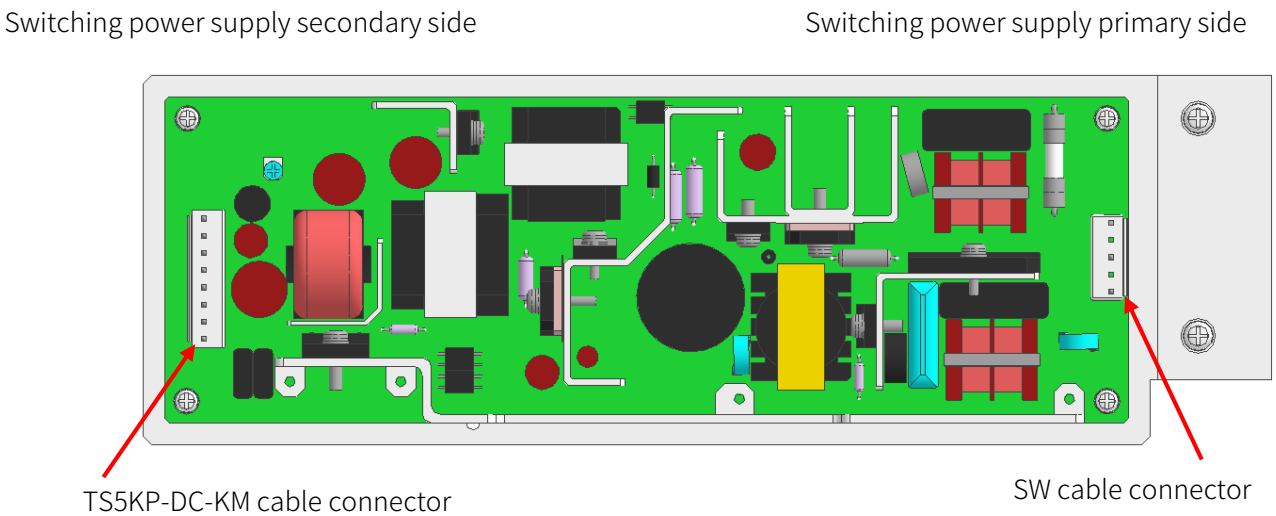


Figure 7.10 Detail of the switching power supply

7.6.6 Fuse replacement

The fuse installed on the front of the controller blows out when the current higher than the specification value flows in the I/O part. If the "I/O Fuse Broken" (8-273) alarm is displayed, the fuse has been blown out. Replace the fuse.

If the fuse of the output module blows out, thoroughly investigate which circuit was in failure, and remove the cause before replacing the fuse.

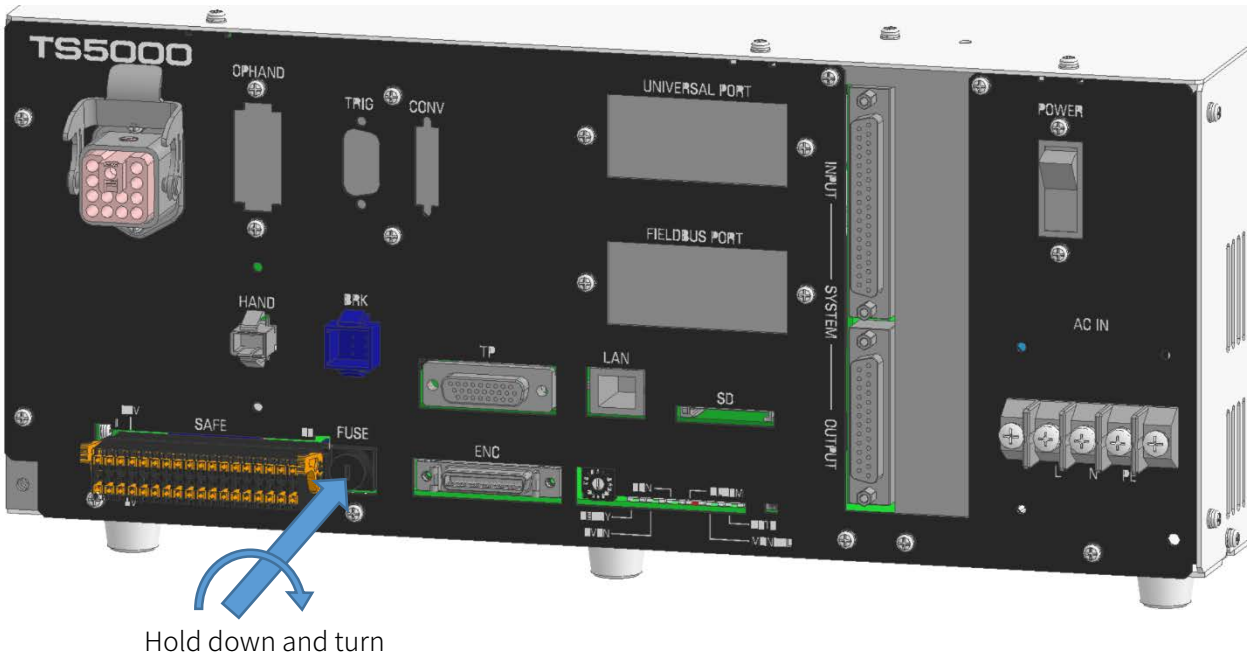


Figure 7.11 Fuse replacement

Table 7.2 Fuse replacement table

Fuse type	Manufacturer
51NM030H	PICO

Replacement procedure

1. **Turn off the P24V power supply connected to the controller on the SAFE connector.**
2. **Use a flat head screwdriver #2 to remove the fuse holder in the figure above. (Turn 90° counter-clockwise while pressing down to unlock.)**
3. **Remove the fuse, and attach a new fuse to the fuse holder.**
4. **Mount the fuse holder. (Turn 90° clockwise while pressing down to lock.)**
5. **Turn on the P24V power supply connected to the controller on the SAFE connector, and confirm that no error occurs.**

8. Replacement Parts for Maintenance

8.1 Precautions Regarding Replacement Parts

⚠DANGER

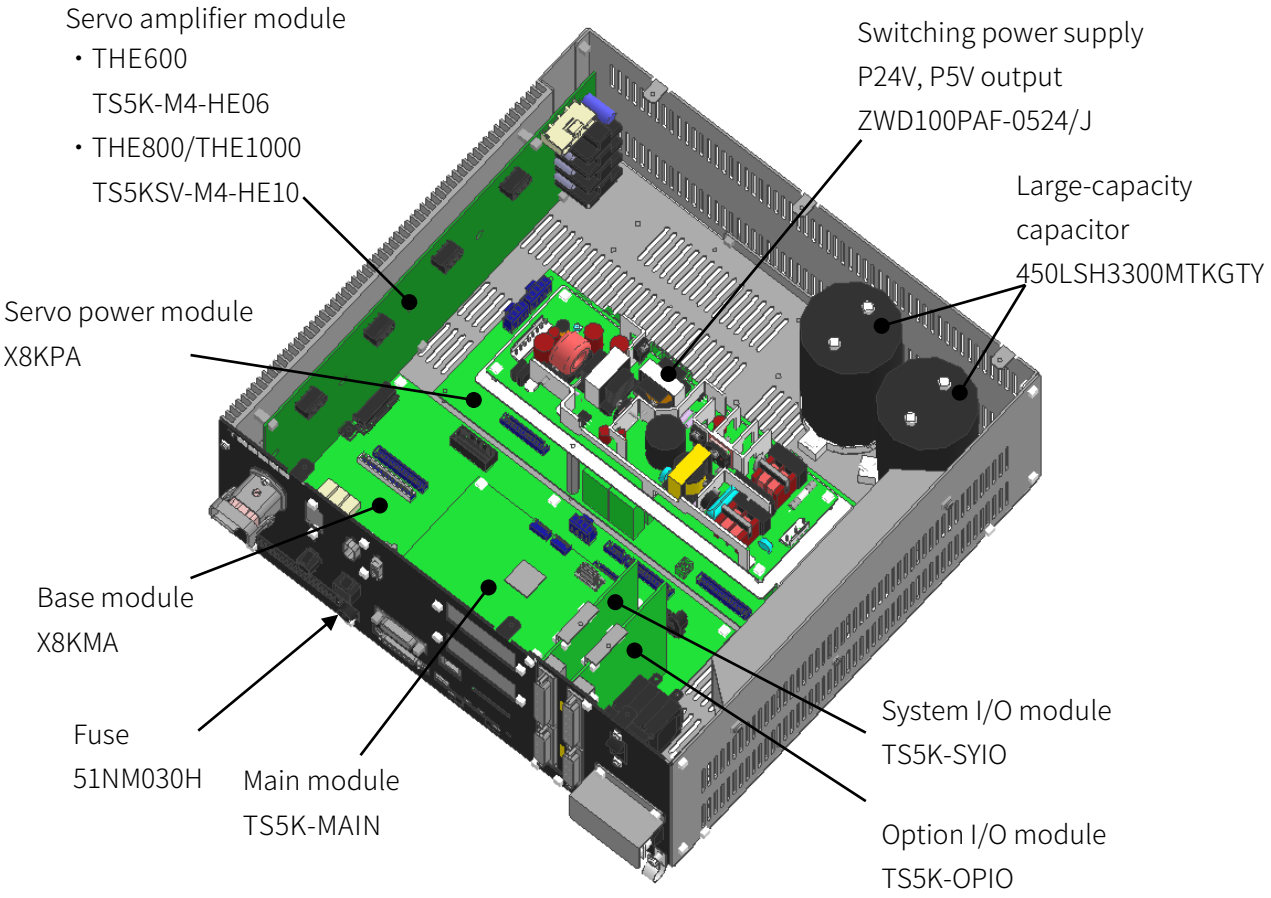
- Always replace with SHIBAURA MACHINE authorized maintenance replacement parts.
  Use of unauthorized parts could result in a fire or other unexpected accident.
  - All parts except for the battery for the position detector are special specifications parts. Please request these parts from SHIBAURA MACHINE when purchasing and ordering.
- Except for replacement of button cell batteries and fuses, please request the replacement work from the SHIBAURA MACHINE servicing company. A fee will be charged for this work.
  SHIBAURA MACHINE will not be liable for any accidents or damage that occurs due to replacement work performed by the customer or its results.

NOTICE

- Always replace with SHIBAURA MACHINE authorized maintenance replacement parts.
  Use of unauthorized parts could result in reduced robot performance and lifespan. A breakdown could also occur.

8.2 Replacement Parts List

No.	Product name	Model	Drawing No.	Manufacturer	Use	Remarks
1	Switching power supply	ZWD100PAF-0524/J	T016683	TDK Corporation	1	P5V, P24V power supply
2	Button cell battery	CR2032	T016686	Maxell Holdings, Ltd.	1	
3	Fuse	51NM030H	T016687	PICO	1	
4	Large-capacity capacitor	450LSH3300MTKGTY		Rubycon Corporation	2	
5	Servo amplifier module (THE600)	TS5K-M4-HE06	L27616G01	SHIBAURA MACHINE CO., LTD	1	
	Servo amplifier module (THE800/THE1000)	TS5KSV-M4-HE10	L27616G02		1	
6	System I/O module	TS5K-SYIO	L27670G01		1	
7	Option I/O module	TS5K-OPIO	L27670G02		1	
8	Main module	TS5K-MAIN	L27669G01		1	
9	Servo power module	X8KPA	L27618G01		1	
10	Base module	X8KMA	L27625G01		1	
11	Teach pendant	TP5000	L27664G01		1	5-m specification


Servo amplifier module
・THE600
TS5K-M4-HE06
・THE800/THE1000
TS5KSV-M4-HE10
Switching power supply
P24V, P5V output
ZWD100PAF-0524/J
Large-capacity
capacitor
450LSH3300MTKGTY
Servo power module
X8KPA
Base module
X8KMA
Fuse
51NM030H
Main module
TS5K-MAIN
System I/O module
TS5K-SYIO
Option I/O module
TS5K-OPIO

8.3 List of Accessories

The whole set of accessories can be purchased with the unit code Y6F0AA1Y0.

No.	Product name	Model	Manufacturer	Use	Remarks
1	I/O jumper connector	M067817	SHIBAURA MACHINE	1	For INPUT
2	TP5000 jumper connector	S981572	SHIBAURA MACHINE	1	For TP
3	Connector	XM2S-2511	OMRON Corporation	1	For OUTPUT Shell
4		XM3A-2521	OMRON Corporation	1	For OUTPUT Plug
5		B2CF 3.50/36/180 LRSNBKBX	Weidmüller Interface GmbH & Co. KG	1	For SAFE
6	SAFE jumper cable	S981311	SHIBAURA MACHINE	4	For SAFE jumper
7	Fuse	51NM030H	PICO	1	External 24 V fuse

Disposal

This explains the disposal of the controller.

9. Disposal

Dispose of this product in compliance with the regulations of the customer. SHIBAURA MACHINE does not provide collection or disposal services.

⚠DANGER

- Never attempt to incinerate, short-circuit, charge, disassemble, or heat the batteries.
  They could rupture and leak, and if the battery fluid gets on your skin or eyes, it will cause inflammation.
  - If battery fluid gets on your skin, immediately flush with water to wash off.
  - If battery fluid gets in your eyes, immediately flush with water to wash off, and then seek medical attention.
  - Dispose of the batteries in compliance with the regulations of the customer.

NOTICE

- Dispose of any unneeded packaging materials in compliance with the regulations of the customer.
  When storing packaging materials, manage them so that they are not used for purposes other than packing the robot.